

Nikodem Boniecki

Nr. albumu: 31865

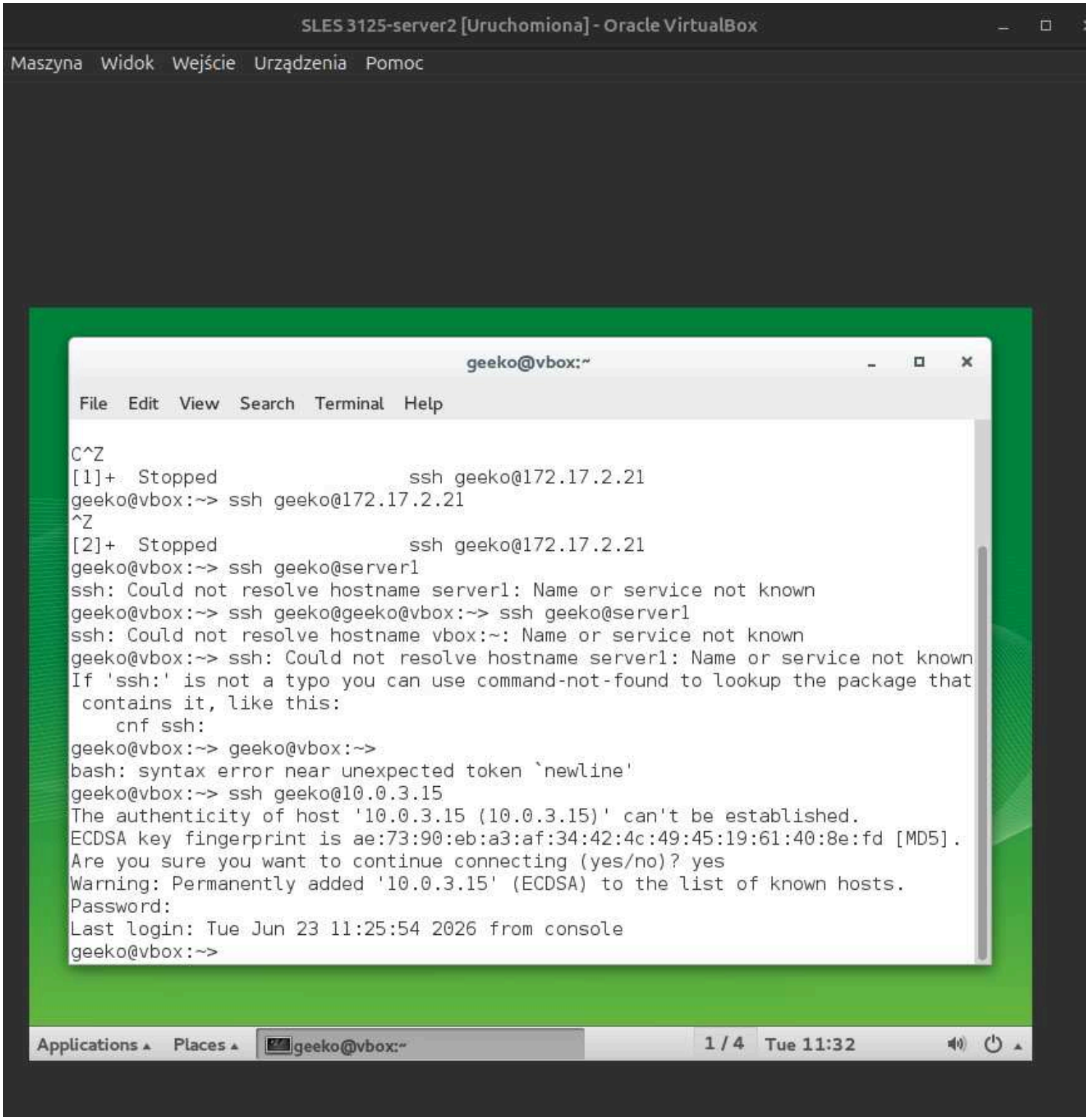
23.06.2026

Lab 1. SSH.

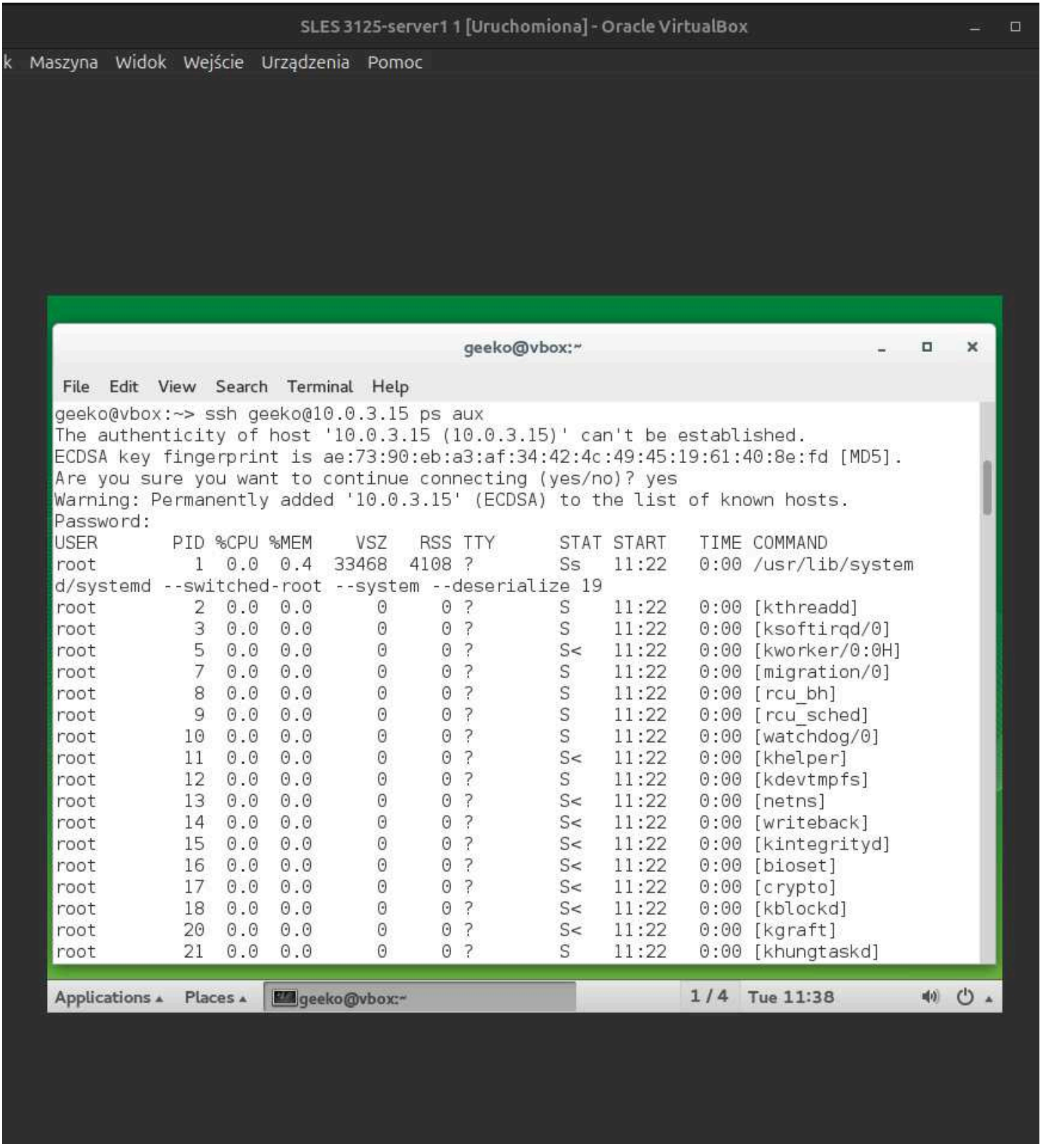
Use ssh to Log Into a Remote Server

Task I

logowanie z server2 na server1



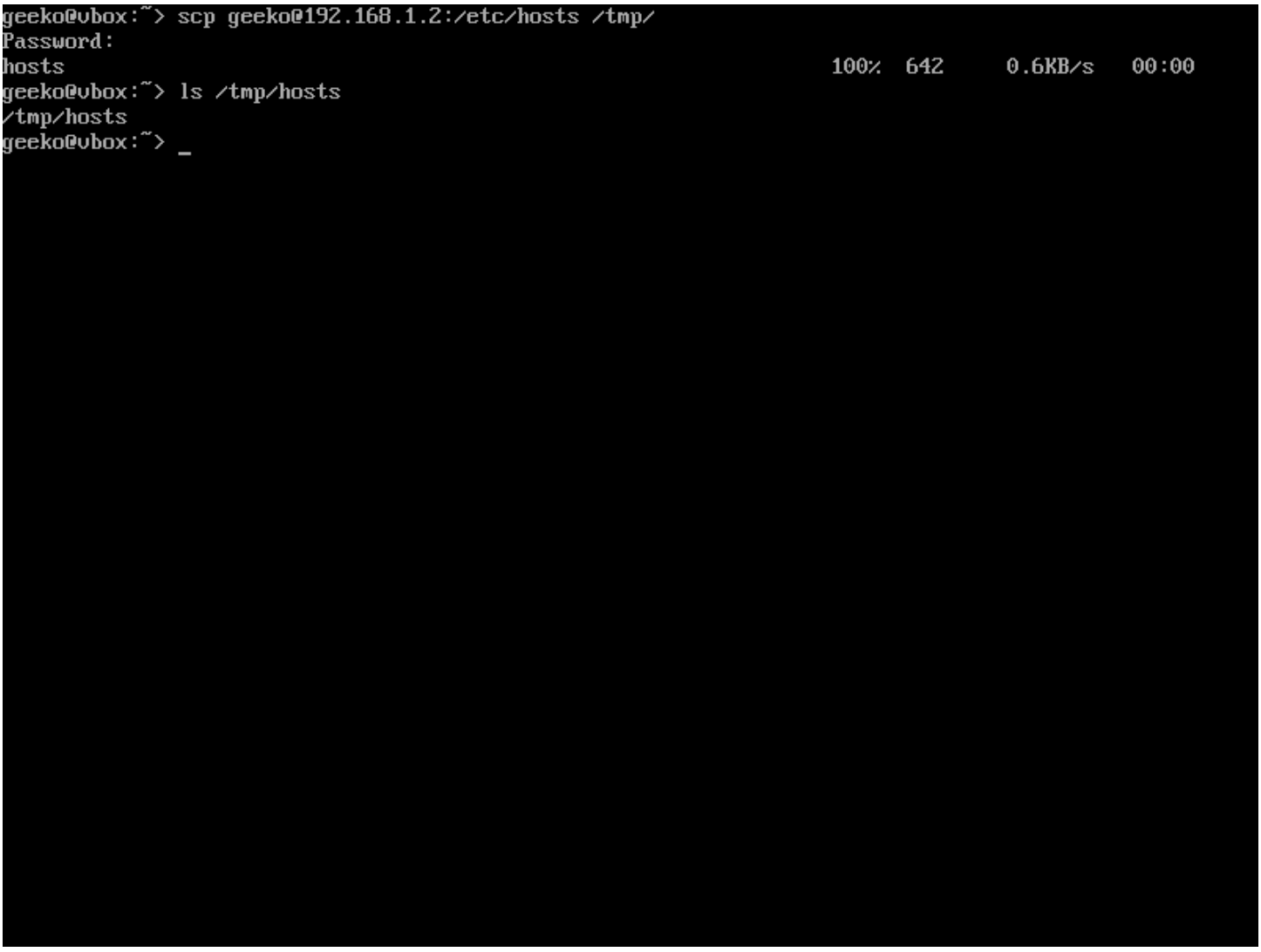
zdalne sprawdzenie procesów



Lab 1. SSH.

Task II

kopiowanie z server2 na server1



wysyłanie pliku z server1



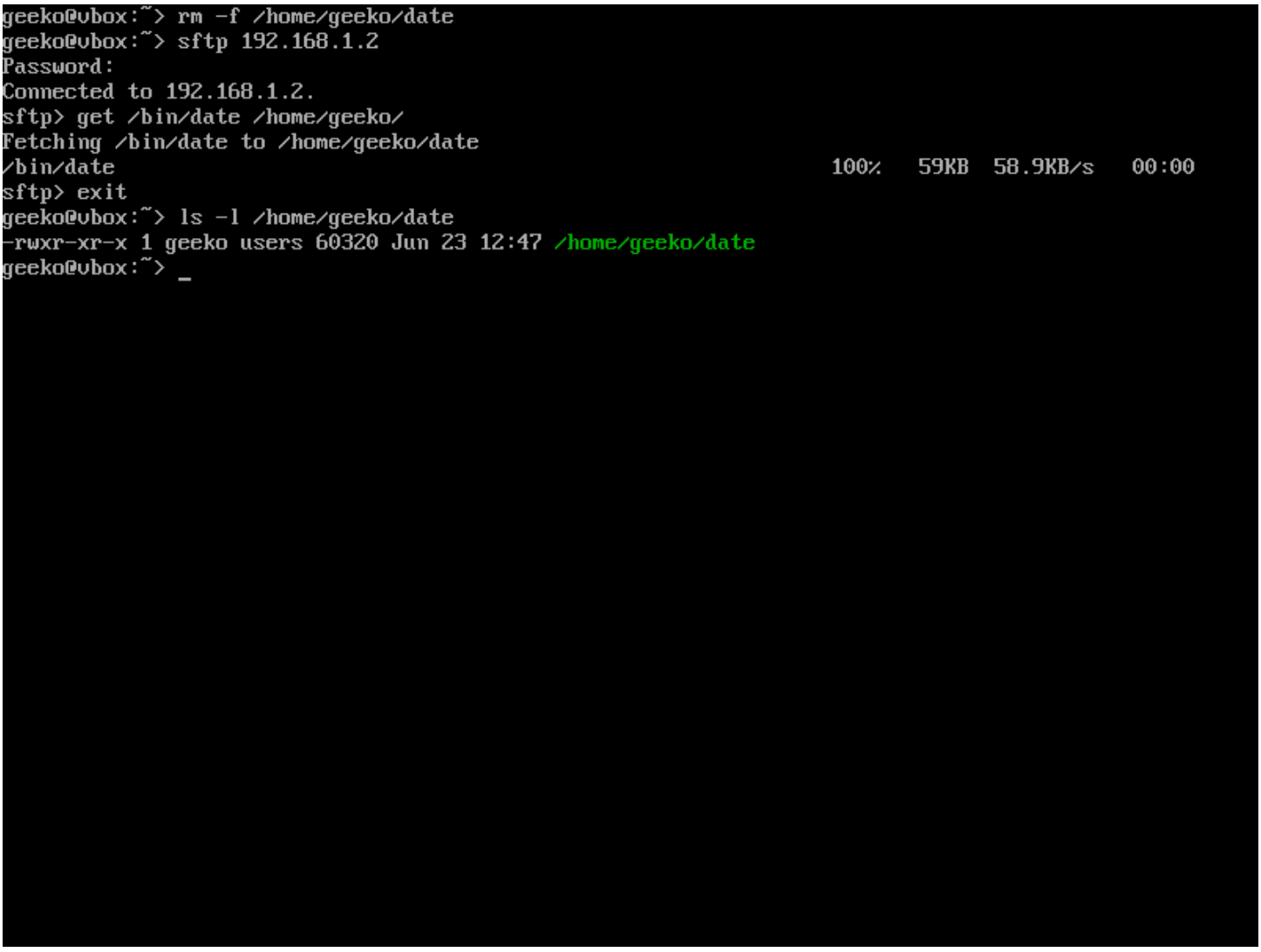
weryfikacja na server2

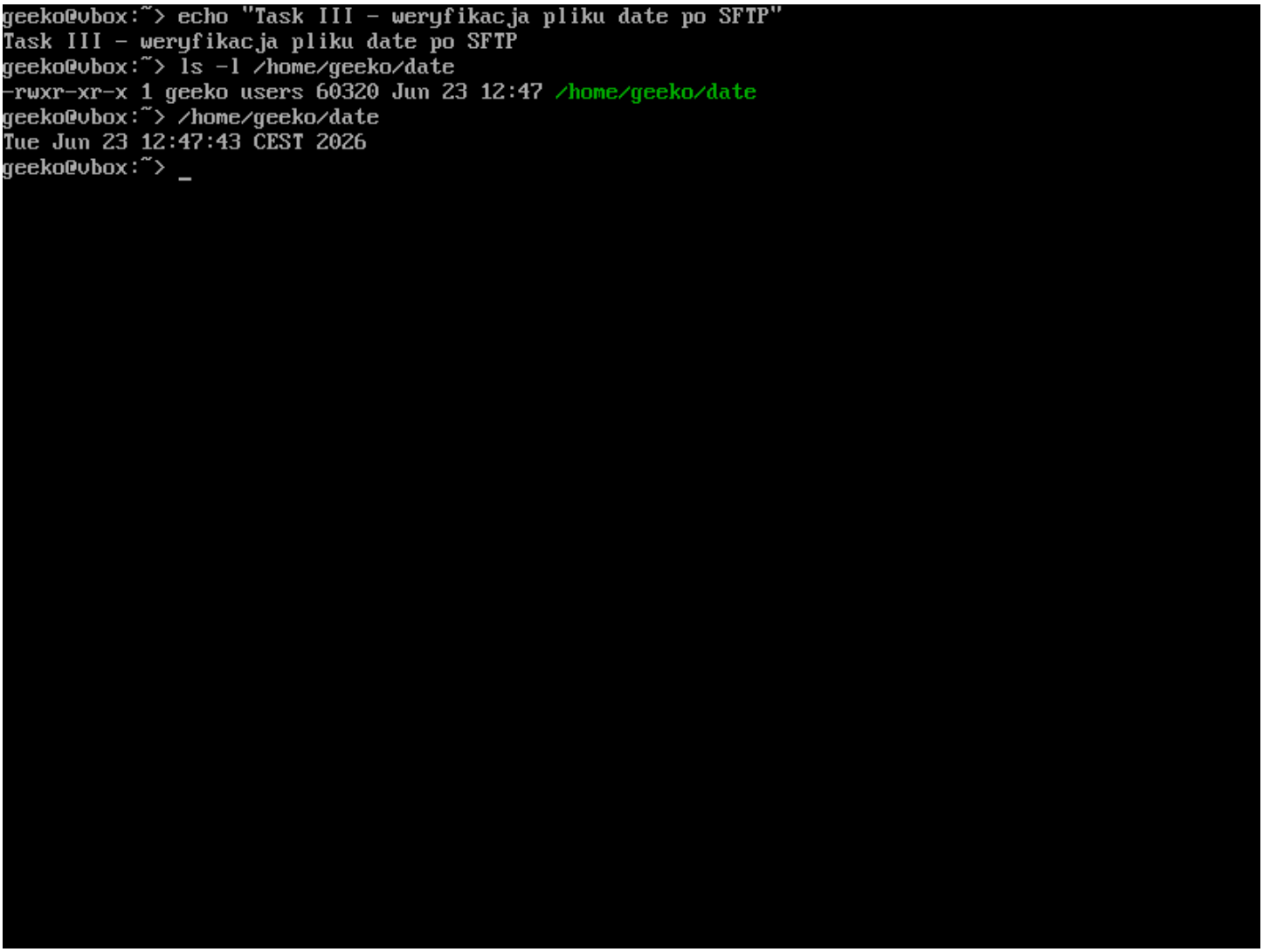
```
geeko@linux-t5co:~> echo "Task II - weryfikacja: plik hosts z Server 1"
Task II - weryfikacja: plik hosts z Server 1
geeko@linux-t5co:~> ls -l /home/geeko/hosts
-rw-r--r-- 1 geeko users 642 Jun 23 12:32 /home/geeko/hosts
geeko@linux-t5co:~> echo linux | sudo -S ip link set eth5 up >/dev/null; true
geeko@linux-t5co:~> ls -l /home/geeko/hosts
-rw-r--r-- 1 geeko users 642 Jun 23 12:34 /home/geeko/hosts
geeko@linux-t5co:~> _
```

Lab 1. SSH.

Task III

kopiowanie plików przez SFTP



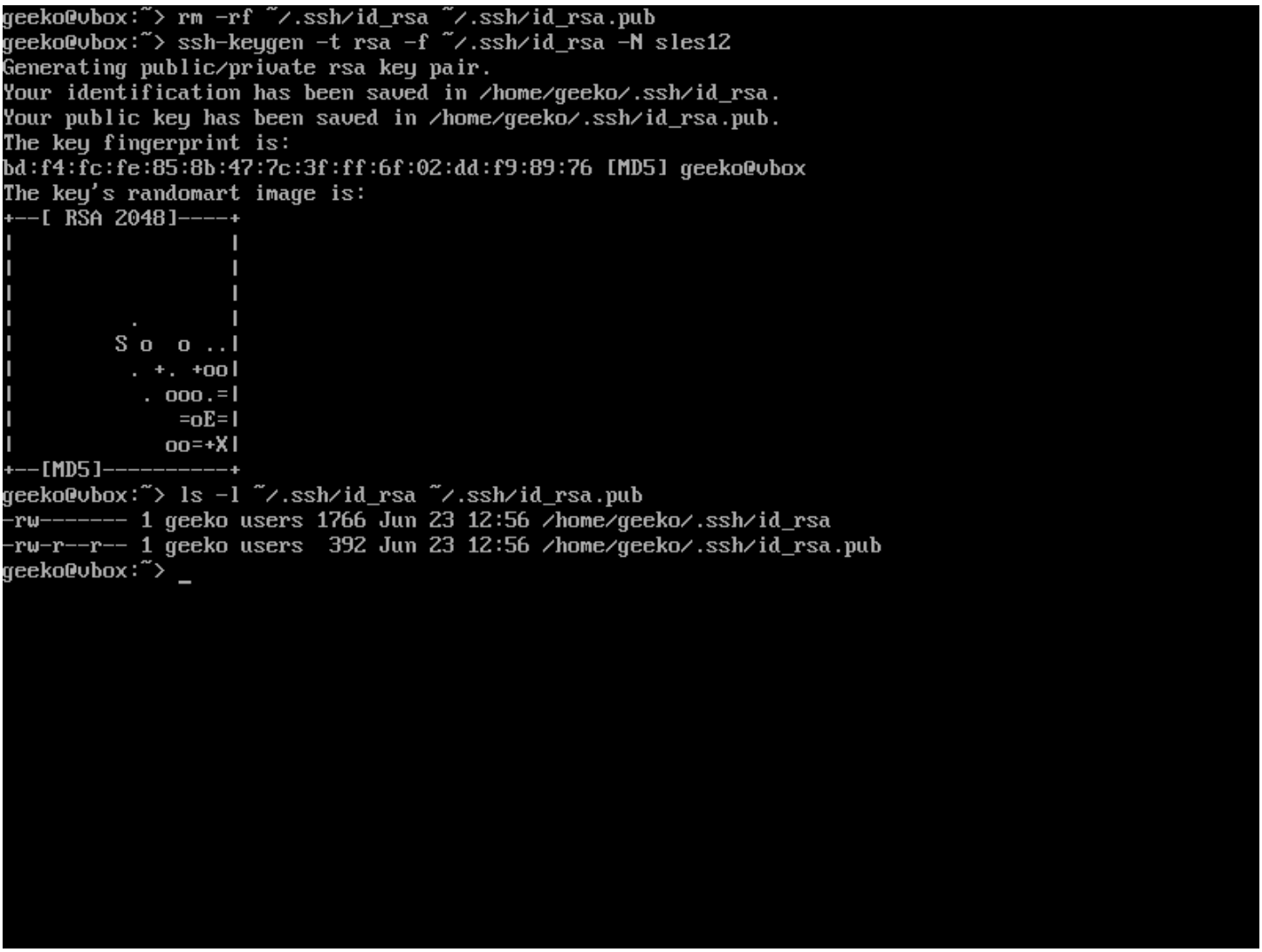

geeko@vbox:~> echo "Task III - weryfikacja pliku date po SFTP"
Task III - weryfikacja pliku date po SFTP
geeko@vbox:~> ls -l /home/geeko/date
-rwxr-xr-x 1 geeko users 60320 Jun 23 12:47 /home/geeko/date
geeko@vbox:~> /home/geeko/date
Tue Jun 23 12:47:43 CEST 2026
geeko@vbox:~> _

Lab 1. SSH.

Configure Key Based Autehntication in OpenSSH

Task I

generowanie pary kluczy RSA



Lab 1. SSH.

Task II

ssh-copy-id na server2

```
geeko@vbox:~> rm -rf ~/.ssh/id_rsa ~/.ssh/id_rsa.pub
geeko@vbox:~> ssh-keygen -t rsa -f ~/.ssh/id_rsa -N sles12
Generating public/private rsa key pair.
Your identification has been saved in /home/geeko/.ssh/id_rsa.
Your public key has been saved in /home/geeko/.ssh/id_rsa.pub.
The key fingerprint is:
bd:f4:fc:fe:85:8b:47:7c:3f:ff:6f:02:dd:f9:89:76 [MD5] geeko@vbox
The key's randomart image is:
+--[ RSA 2048]----+
|                 |
|                 |
|                 |
|                 |
|        .        |
|       S o  o ..|
|         . +. +oo|
|          . ooo.=|
|             =oE=|
|            oo=+X|
+--[MD5]----------+
geeko@vbox:~> ls -l ~/.ssh/id_rsa ~/.ssh/id_rsa.pub
-rw------- 1 geeko users 1766 Jun 23 12:56 /home/geeko/.ssh/id_rsa
-rw-r--r-- 1 geeko users  392 Jun 23 12:56 /home/geeko/.ssh/id_rsa.pub
geeko@vbox:~> ssh-copy-id -i ~/.ssh/id_rsa.pub 192.168.1.2
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are alr
eady installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to inst
all the new keys
Password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh '192.168.1.2'"
and check to make sure that only the key(s) you wanted were added.

geeko@vbox:~>
```

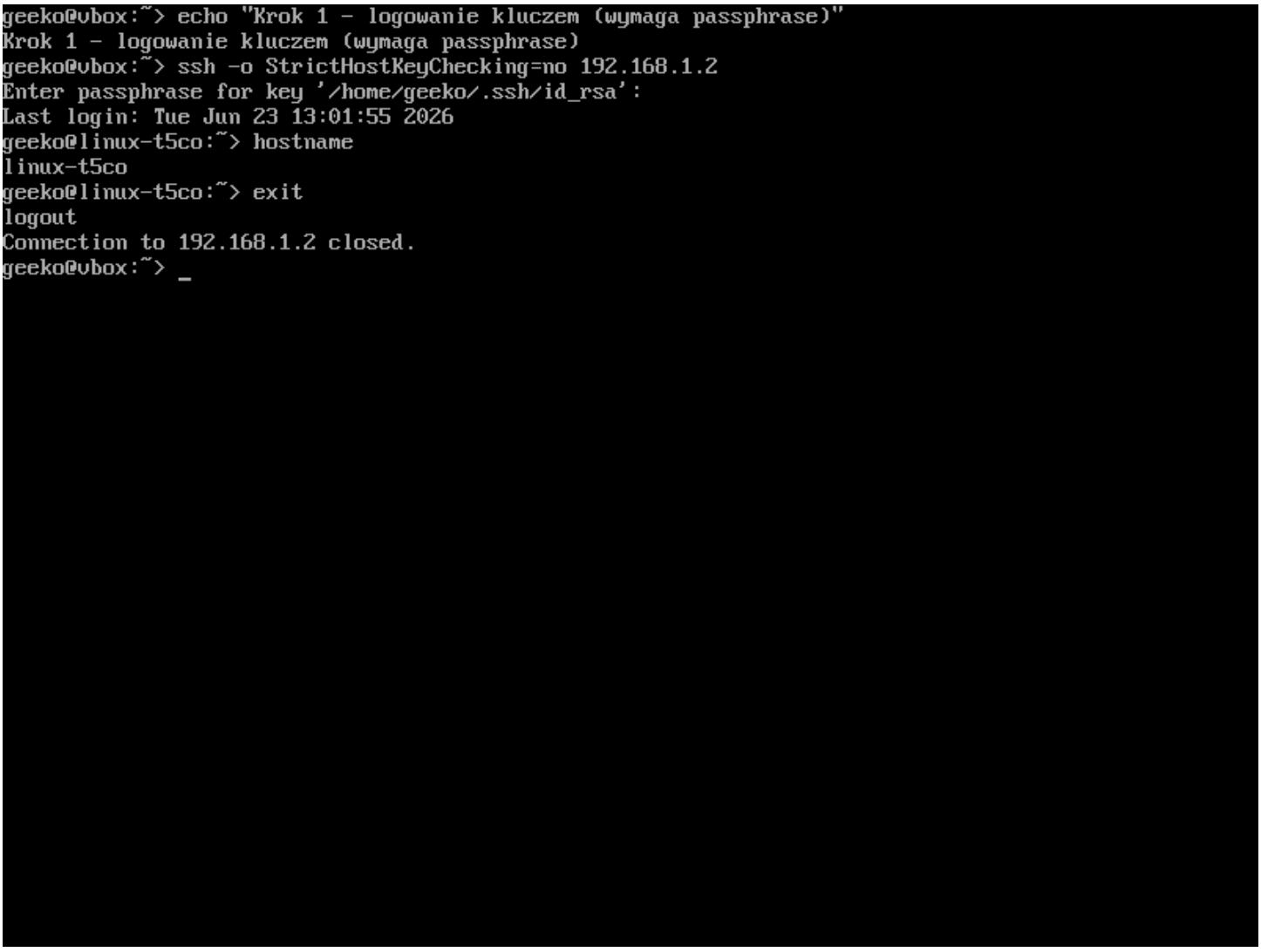
test logowania kluczem

```
geeko@vbox:~> echo "Task II - test logowania kluczem (passphrase: sles12)"
Task II - test logowania kluczem (passphrase: sles12)
geeko@vbox:~> ssh -o StrictHostKeyChecking=no 192.168.1.2 hostname
Enter passphrase for key '/home/geeko/.ssh/id_rsa':
linux-t5co
geeko@vbox:~> _
```

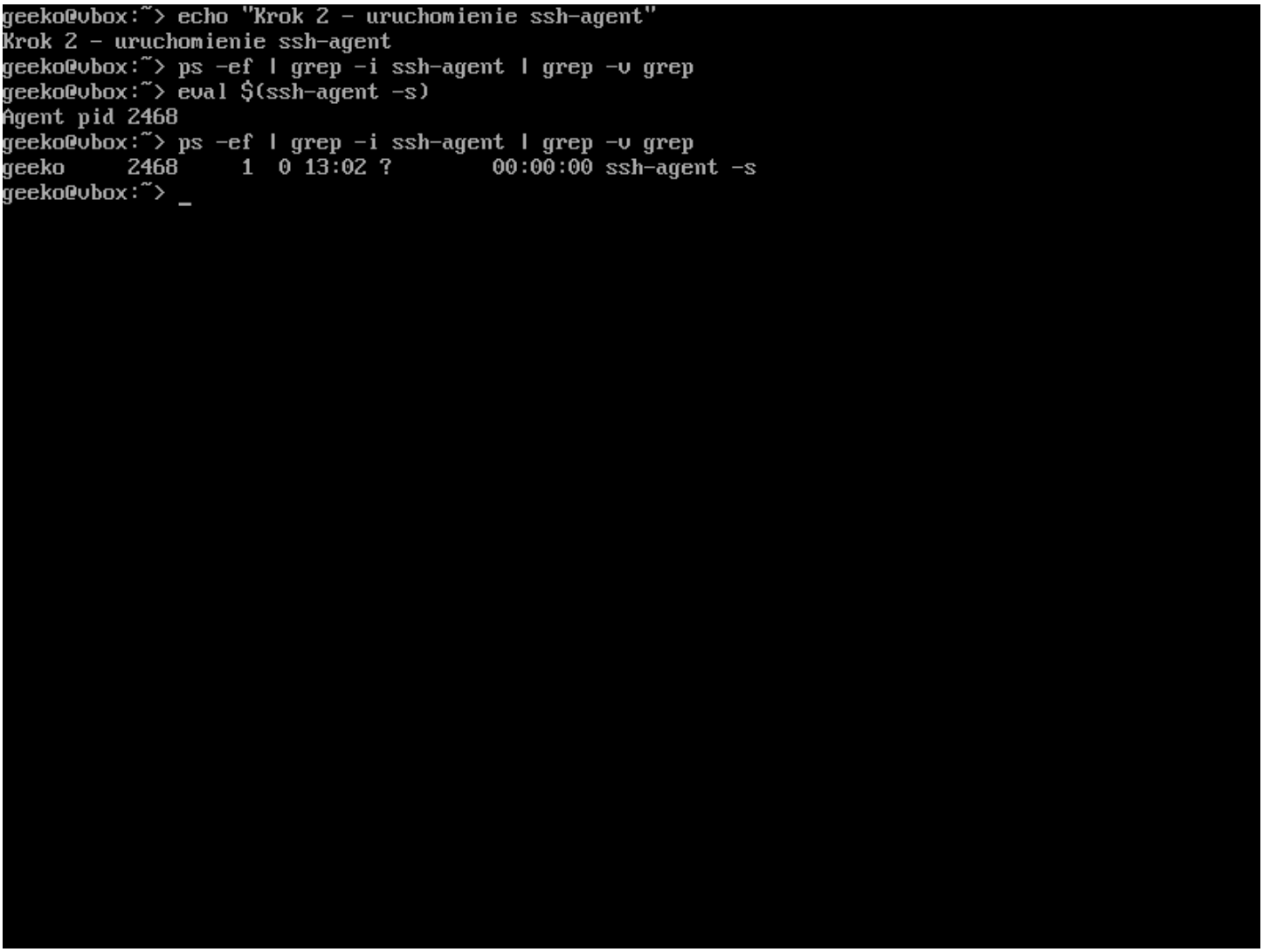
Lab 1. SSH.

Task III

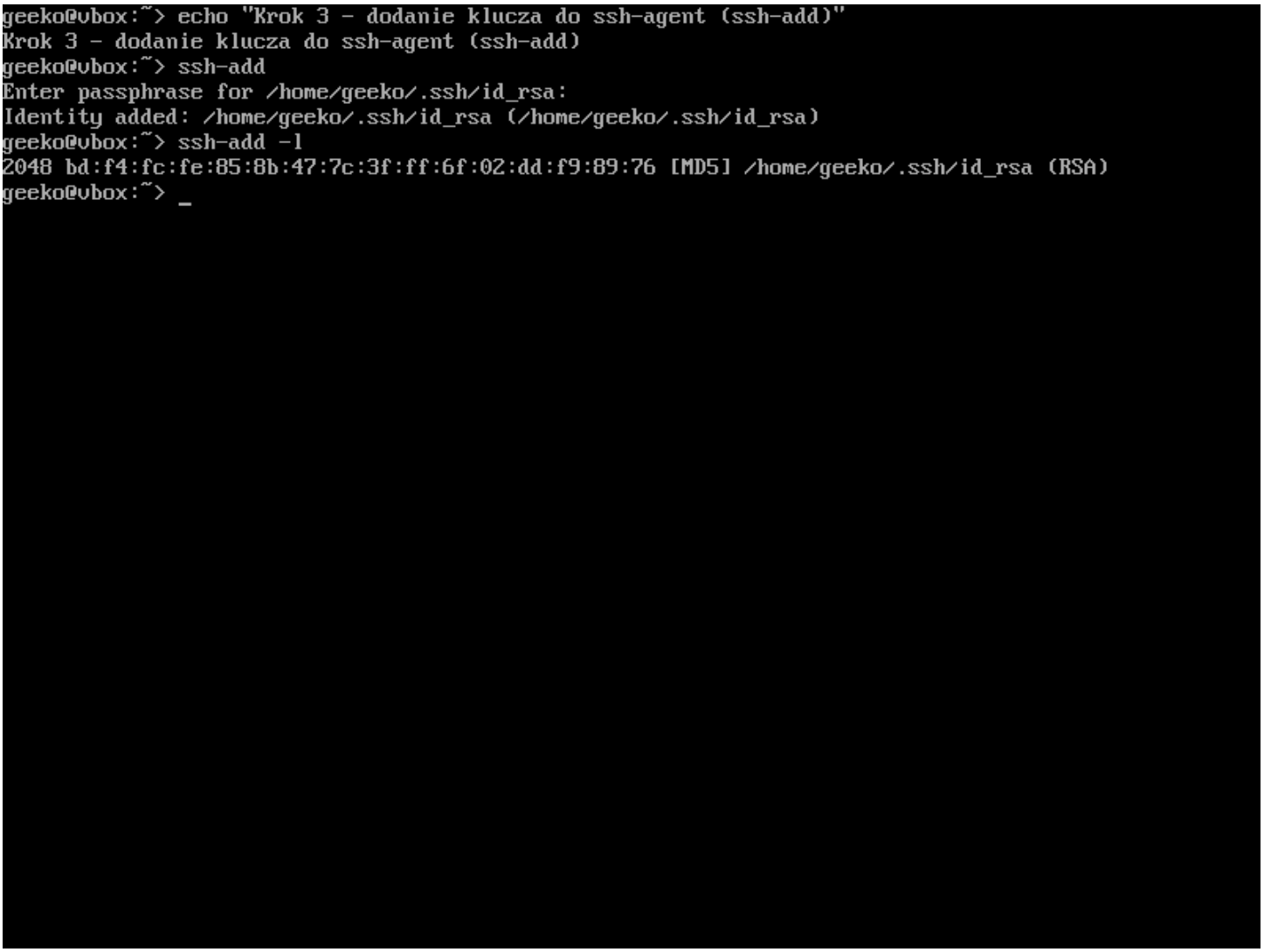
logowanie ssh 192.168.1.2 z prośbą o passphrase sles12



sprawdzenie i uruchomienie ssh-agent


geeko@vbox:~> echo "Krok 2 - uruchomienie ssh-agent"
Krok 2 - uruchomienie ssh-agent
geeko@vbox:~> ps -ef | grep -i ssh-agent | grep -v grep
geeko@vbox:~> eval $(ssh-agent -s)
Agent pid 2468
geeko@vbox:~> ps -ef | grep -i ssh-agent | grep -v grep
geeko 2468 1 0 13:02 ? 00:00:00 ssh-agent -s
geeko@vbox:~> _

ssh-add



logowanie bez passpharase


geeko@vbox:~> echo "Krok 4 - logowanie bez passphrase (ssh-agent)"
Krok 4 - logowanie bez passphrase (ssh-agent)
geeko@vbox:~> ssh -o StrictHostKeyChecking=no 192.168.1.2 hostname
linux-t5co
geeko@vbox:~> _

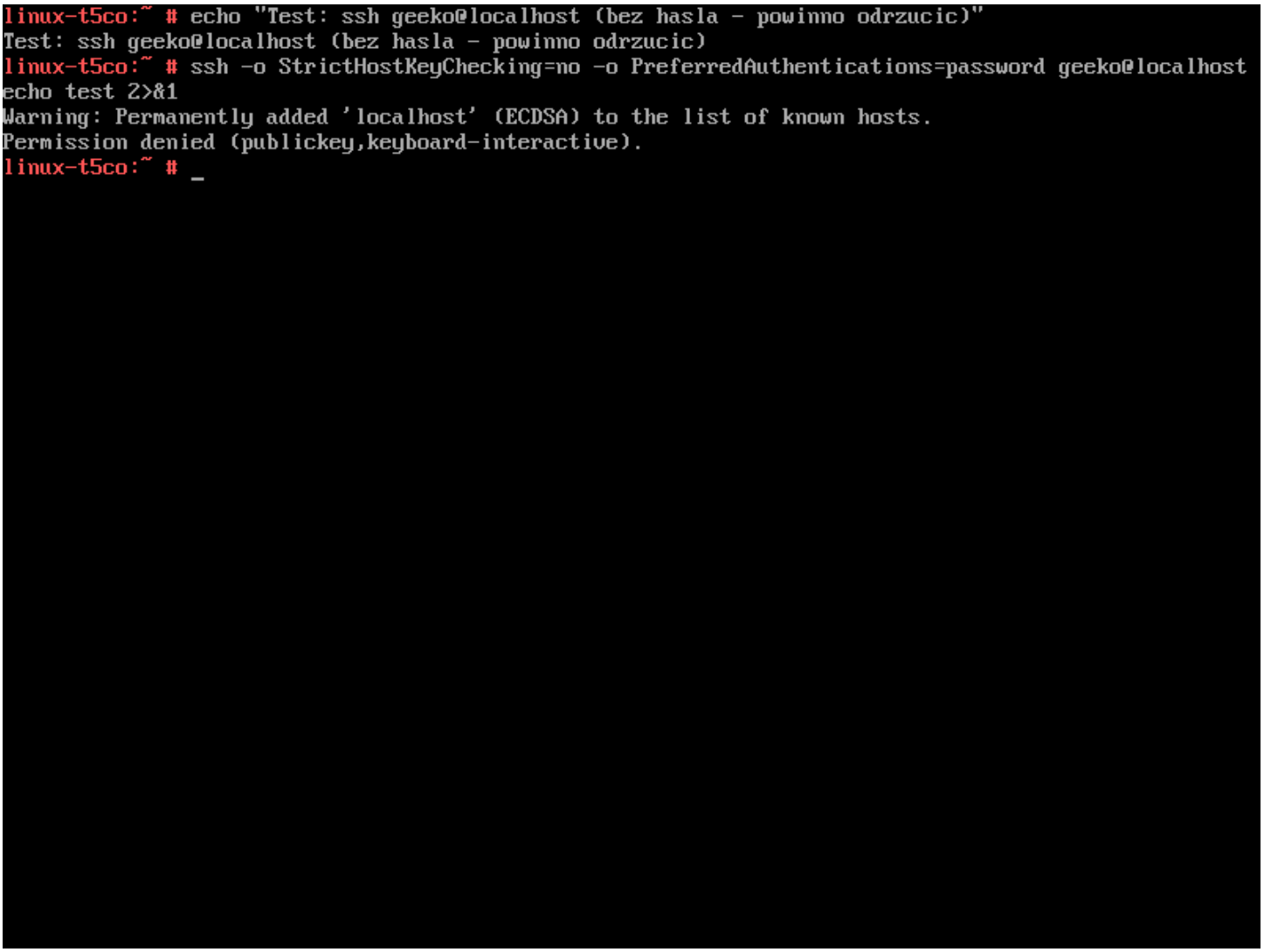
Lab 1. SSH.

Task IV

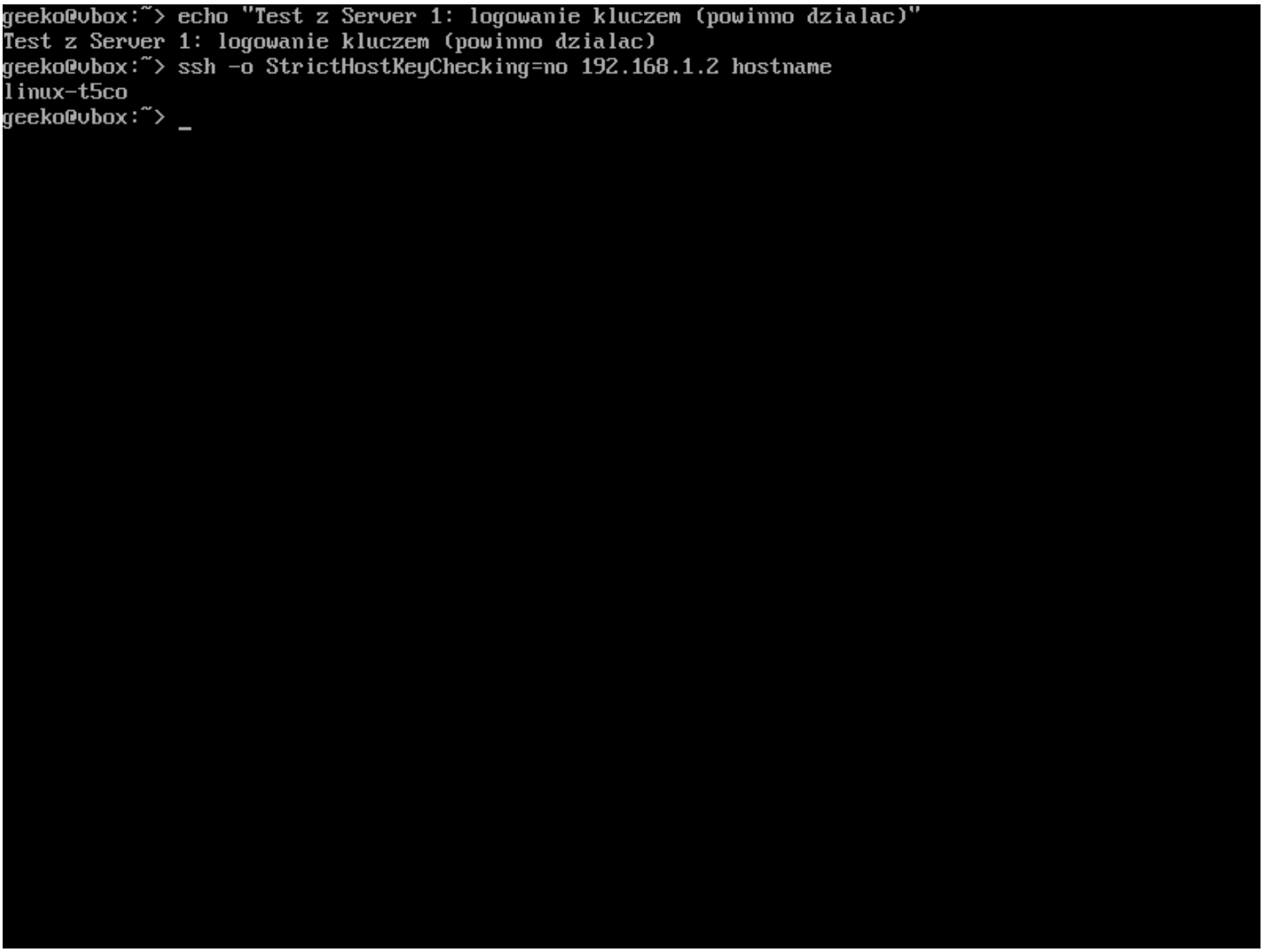
edycja sshd_config

```
geeko@linux-t5co:~> echo "Task IV - wylaczenie logowania haslem na Server 2"
Task IV - wylaczenie logowania haslem na Server 2
geeko@linux-t5co:~> su -
Password:
linux-t5co:~ # sed -i 's/^#*PasswordAuthentication.*/PasswordAuthentication no/' /etc/ssh/sshd_confi
g
linux-t5co:~ # su -
linux-t5co:~ # linux
If 'linux' is not a typo you can use command-not-found to lookup the package that contains it, like
this:
    cnf linux
linux-t5co:~ # sed -i 's/^#*UsePAM.*/UsePAM no/' /etc/ssh/sshd_config
linux-t5co:~ # su -
linux-t5co:~ # linux
If 'linux' is not a typo you can use command-not-found to lookup the package that contains it, like
this:
    cnf linux
linux-t5co:~ # grep -E '^(PasswordAuthentication|UsePAM)' /etc/ssh/sshd_config
PasswordAuthentication no
UsePAM no
linux-t5co:~ # su -
linux-t5co:~ # linux
If 'linux' is not a typo you can use command-not-found to lookup the package that contains it, like
this:
    cnf linux
linux-t5co:~ # systemctl restart sshd
linux-t5co:~ # exit
logout
linux-t5co:~ # _
```

test ssh - odrzucenie



z server1 log. kluczem - działa



cofam zmiany

```
linux-t5co:~ # echo "Cofniecie zmian - przywrocenie PasswordAuthentication yes"
Cofniecie zmian - przywrocenie PasswordAuthentication yes
linux-t5co:~ # su -
linux-t5co:~ # linux
If 'linux' is not a typo you can use command-not-found to lookup the package that contains it, like
this:
    cnf linux
linux-t5co:~ # sed -i 's/^#*PasswordAuthentication.*/PasswordAuthentication yes/' /etc/ssh/sshd_conf
ig
linux-t5co:~ # su -
linux-t5co:~ # linux
If 'linux' is not a typo you can use command-not-found to lookup the package that contains it, like
this:
    cnf linux
linux-t5co:~ # sed -i 's/^#*UsePAM.*/UsePAM yes/' /etc/ssh/sshd_config
linux-t5co:~ # su -
linux-t5co:~ # linux
If 'linux' is not a typo you can use command-not-found to lookup the package that contains it, like
this:
    cnf linux
linux-t5co:~ # grep -E '^(PasswordAuthentication|UsePAM)' /etc/ssh/sshd_config
PasswordAuthentication yes
UsePAM yes
linux-t5co:~ # su -
linux-t5co:~ # linux
If 'linux' is not a typo you can use command-not-found to lookup the package that contains it, like
this:
    cnf linux
linux-t5co:~ # systemctl restart sshd
linux-t5co:~ # exit
logout
linux-t5co:~ #
```

Lab 1. SSH.

1.4 Stwórz skrypt na server1 na koncie geeko.

skrypt

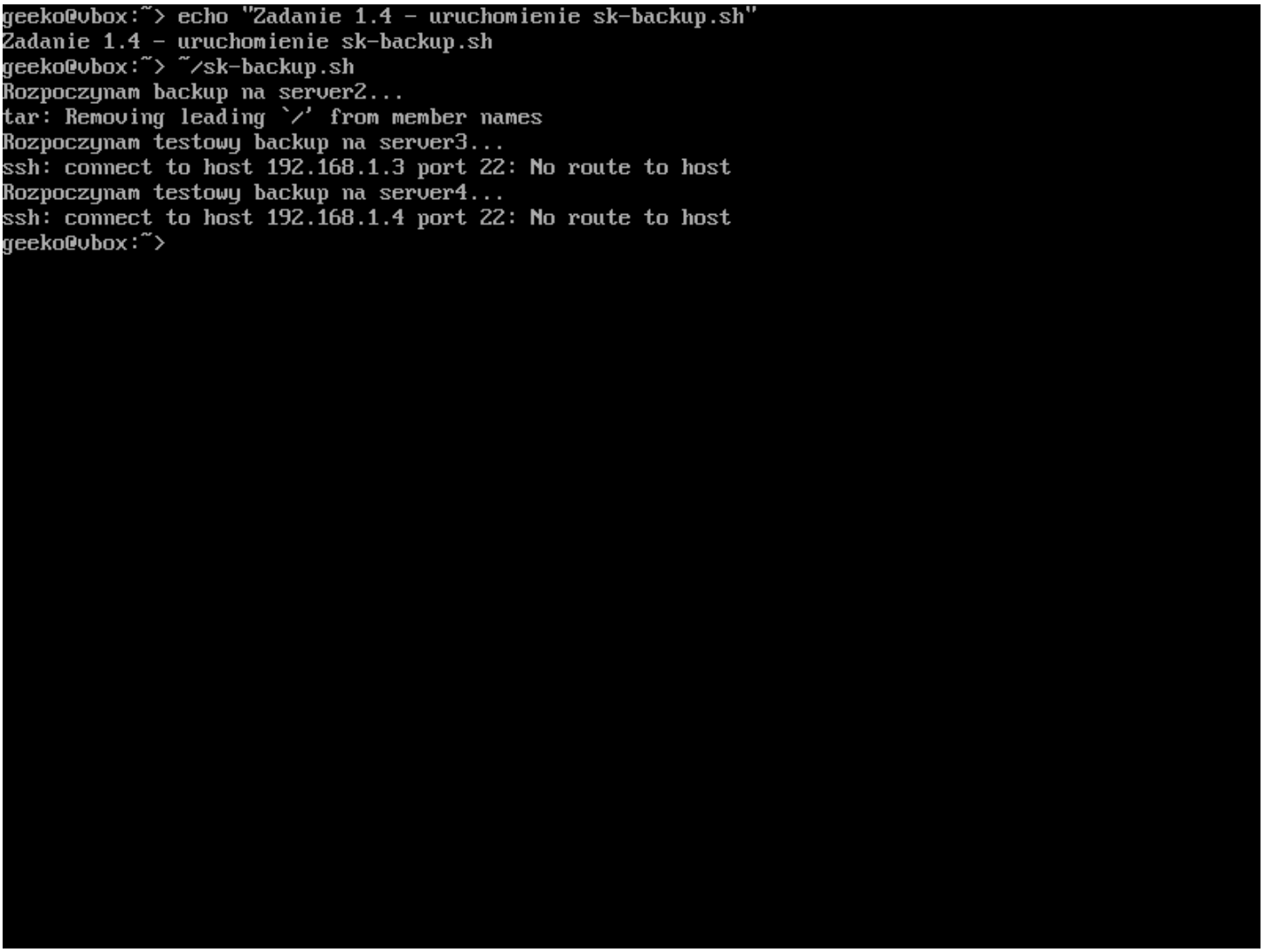
```
geeko@vbox:~> echo "Zadanie 1.4 - konfiguracja klucza geeko -> root@server2"
Zadanie 1.4 - konfiguracja klucza geeko -> root@server2
geeko@vbox:~> ssh-copy-id -i ~/.ssh/id_rsa.pub root@192.168.1.2
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are alr
eady installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to inst
all the new keys
Password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'root@192.168.1.2'"
and check to make sure that only the key(s) you wanted were added.

geeko@vbox:~> rm -f ~/sk-backup.sh
geeko@vbox:~> echo '#!/bin/bash' >> ~/sk-backup.sh
geeko@vbox:~> echo 'echo "Rozpoczynam backup na server2..."' >> ~/sk-backup.sh
geeko@vbox:~> echo 'ssh -o StrictHostKeyChecking=no root@192.168.1.2 "tar -czf /backup-home.tgz /hom
e"' >> ~/sk-backup.sh
geeko@vbox:~> echo 'echo "Rozpoczynam testowy backup na server3..."' >> ~/sk-backup.sh
geeko@vbox:~> echo 'ssh -o StrictHostKeyChecking=no root@192.168.1.3 "tar -czf /backup-home.tgz /hom
e"' >> ~/sk-backup.sh
geeko@vbox:~> echo 'echo "Rozpoczynam testowy backup na server4..."' >> ~/sk-backup.sh
geeko@vbox:~> echo 'ssh -o StrictHostKeyChecking=no root@192.168.1.4 "tar -czf /backup-home.tgz /hom
e"' >> ~/sk-backup.sh
geeko@vbox:~> chmod +x ~/sk-backup.sh
geeko@vbox:~> cat ~/sk-backup.sh
#!/bin/bash
echo "Rozpoczynam backup na server2..."
ssh -o StrictHostKeyChecking=no root@192.168.1.2 "tar -czf /backup-home.tgz /home"
echo "Rozpoczynam testowy backup na server3..."
ssh -o StrictHostKeyChecking=no root@192.168.1.3 "tar -czf /backup-home.tgz /home"
echo "Rozpoczynam testowy backup na server4..."
ssh -o StrictHostKeyChecking=no root@192.168.1.4 "tar -czf /backup-home.tgz /home"
geeko@vbox:~> _
```

działanie



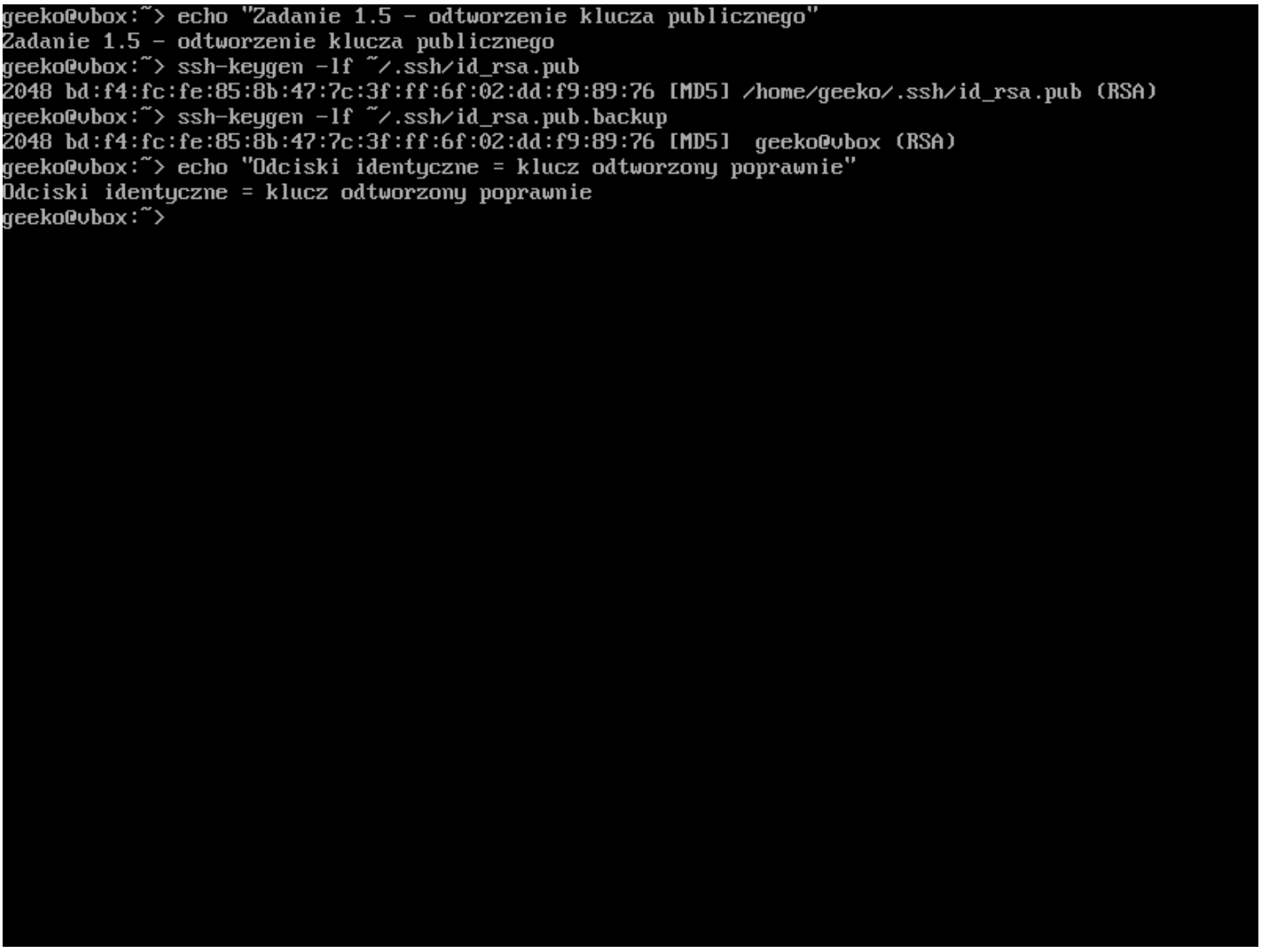
backup sprawdzenie

```
linux-t5co:~ # echo "Weryfikacja: plik backup na server2"
Weryfikacja: plik backup na server2
linux-t5co:~ # ls -lh /backup-home.tgz
-rw-r--r-- 1 root root 412K Jun 23 13:17 /backup-home.tgz
linux-t5co:~ # tar -tzf /backup-home.tgz | head -5
home/
home/geeko/
home/geeko/.config/
home/geeko/.config/user-dirs.dirs
home/geeko/.config/user-dirs.locale
linux-t5co:~ #
```

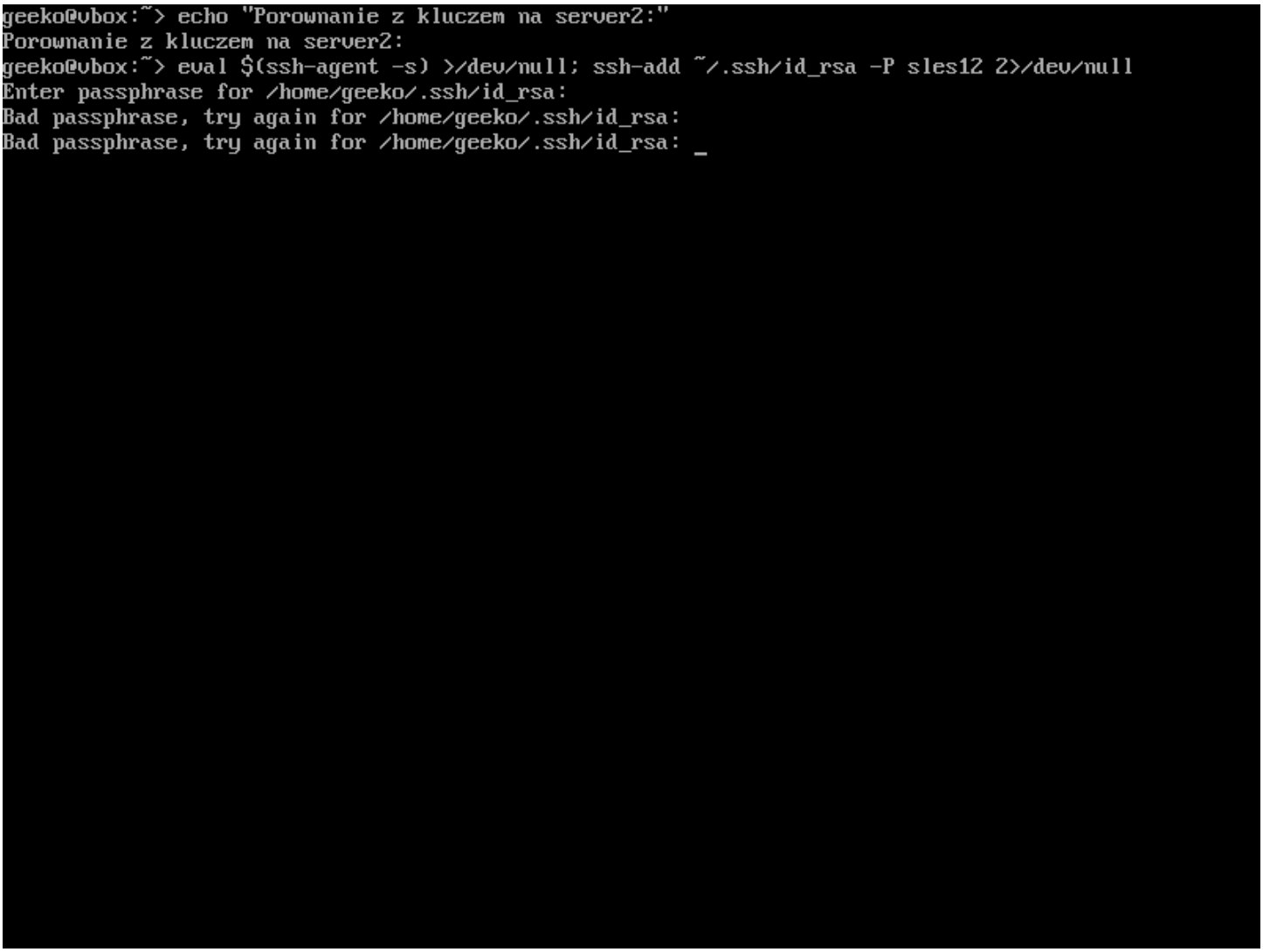
Lab 1. SSH.

1.5. Generowanie klucza publicznego na podstawie prywatnego.

identyczne odciski



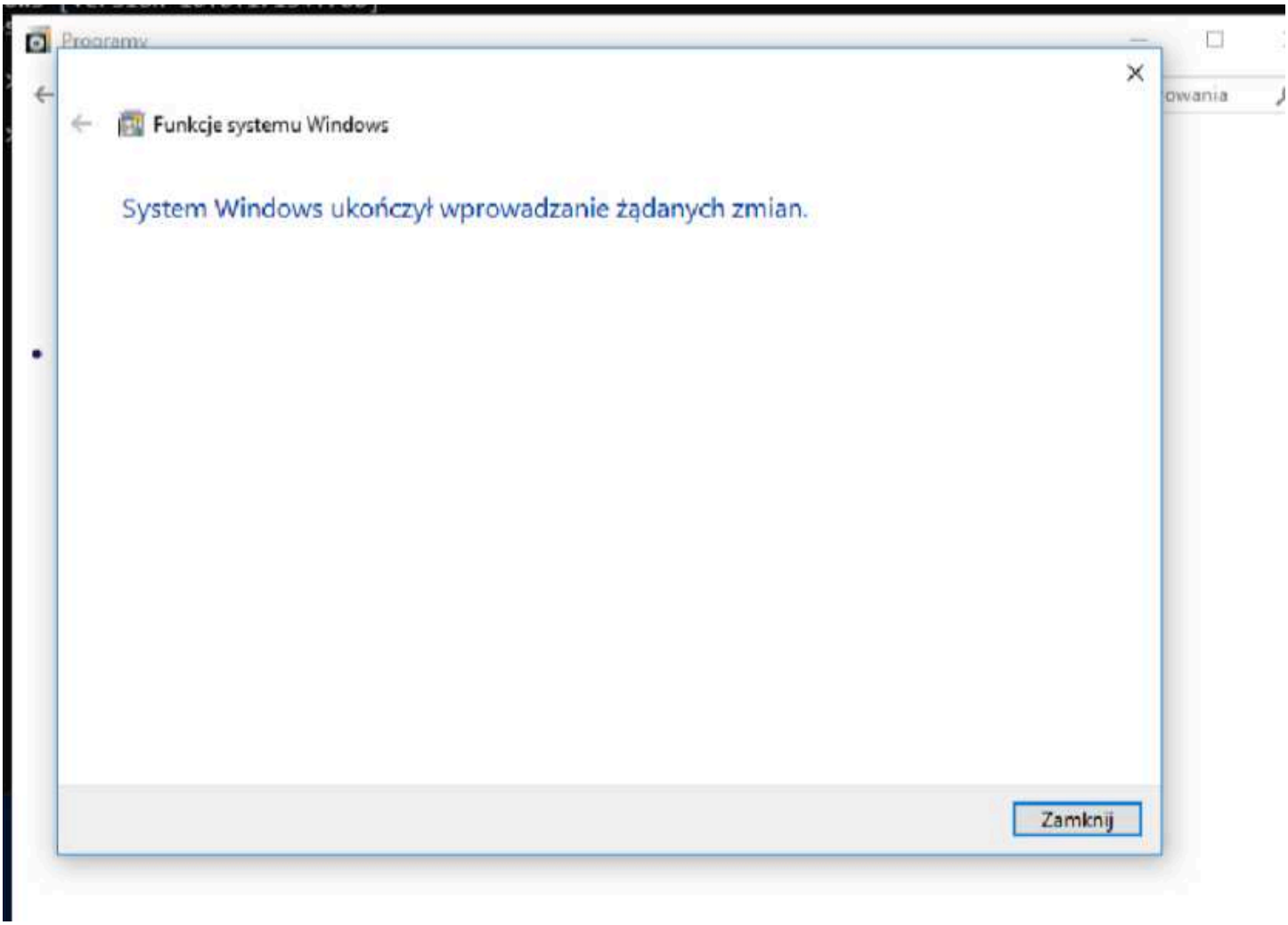
porównanie klucza z server2


geeko@vbox:~> echo "Porownanie z kluczem na server2:"
Porownanie z kluczem na server2:
geeko@vbox:~> eval $(ssh-agent -s) >/dev/null; ssh-add ~/.ssh/id_rsa -P sles12 2>/dev/null
Enter passphrase for /home/geeko/.ssh/id_rsa:
Bad passphrase, try again for /home/geeko/.ssh/id_rsa:
Bad passphrase, try again for /home/geeko/.ssh/id_rsa:

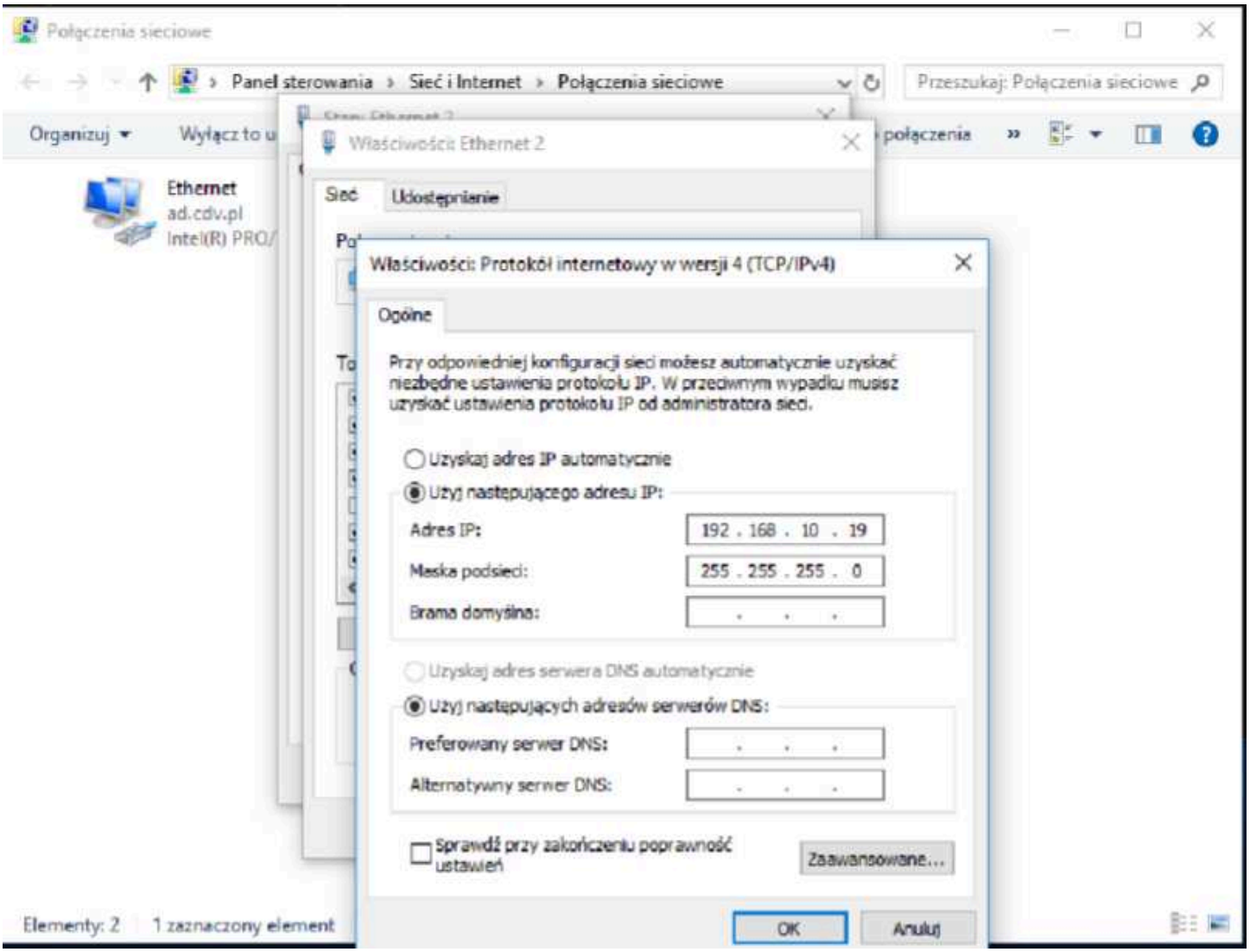
symulacja

```
geeko@vbox:~> echo "Zadanie 1.5 - symulacja utraty klucza publicznego"
Zadanie 1.5 - symulacja utraty klucza publicznego
geeko@vbox:~> cp ~/.ssh/id_rsa.pub ~/.ssh/id_rsa.pub.backup
geeko@vbox:~> rm ~/.ssh/id_rsa.pub
geeko@vbox:~> ls ~/.ssh/id_rsa.pub 2>&1
ls: cannot access /home/geeko/.ssh/id_rsa.pub: No such file or directory
geeko@vbox:~> ssh-keygen -y -f ~/.ssh/id_rsa -P sles12 > ~/.ssh/id_rsa.pub
geeko@vbox:~> echo "Porownanie z kopia zapasowa:"
Porownanie z kopia zapasowa:
geeko@vbox:~> diff ~/.ssh/id_rsa.pub ~/.ssh/id_rsa.pub.backup && echo "Klucze sa IDENTYCZNE"
1c1
< ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQC+VOoCWL4CxWhwlOqxZaLeKgXFI1FeIklE5d98yO2gdZSBhrpCykUOm2GbYd
JivuTeDYZGgl9k0f/6Gz0IJxcS7eKOrtxBYvFkG5d69qBLk0w54thKLaNFv64zsU4IEWJw1clepuiiEd2jwxfWztV/tfolvJnFww
cgjbV+UqyAS1ZcFVVf9mNhic2TSBS2xeOYej3wFNgarq3JuQc894WATQKAMW2Y5AZbzfyG4LU8+lRiHZNBRrysw1BVFuj/oKS/PO
7S6EDOVNXLvFjrCgpTJKavmKrz/WyWnk3ZxpNT1z+XE0r//dgeo7hvhQdBE2KToZ7H0YihxWNxCFj5gAfD
---
> ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQC+VOoCWL4CxWhwlOqxZaLeKgXFI1FeIklE5d98yO2gdZSBhrpCykUOm2GbYd
JivuTeDYZGgl9k0f/6Gz0IJxcS7eKOrtxBYvFkG5d69qBLk0w54thKLaNFv64zsU4IEWJw1clepuiiEd2jwxfWztV/tfolvJnFww
cgjbV+UqyAS1ZcFVVf9mNhic2TSBS2xeOYej3wFNgarq3JuQc894WATQKAMW2Y5AZbzfyG4LU8+lRiHZNBRrysw1BVFuj/oKS/PO
7S6EDOVNXLvFjrCgpTJKavmKrz/WyWnk3ZxpNT1z+XE0r//dgeo7hvhQdBE2KToZ7H0YihxWNxCFj5gAfD geeko@vbox
geeko@vbox:~> md5sum ~/.ssh/id_rsa.pub ~/.ssh/id_rsa.pub.backup
35c2cf98e5f36569a107e3928187467e  /home/geeko/.ssh/id_rsa.pub
5a8989064f125dfd55d2c00ec39aa996  /home/geeko/.ssh/id_rsa.pub.backup
geeko@vbox:~> _
```

Lab 2. OpenSSL - PKI

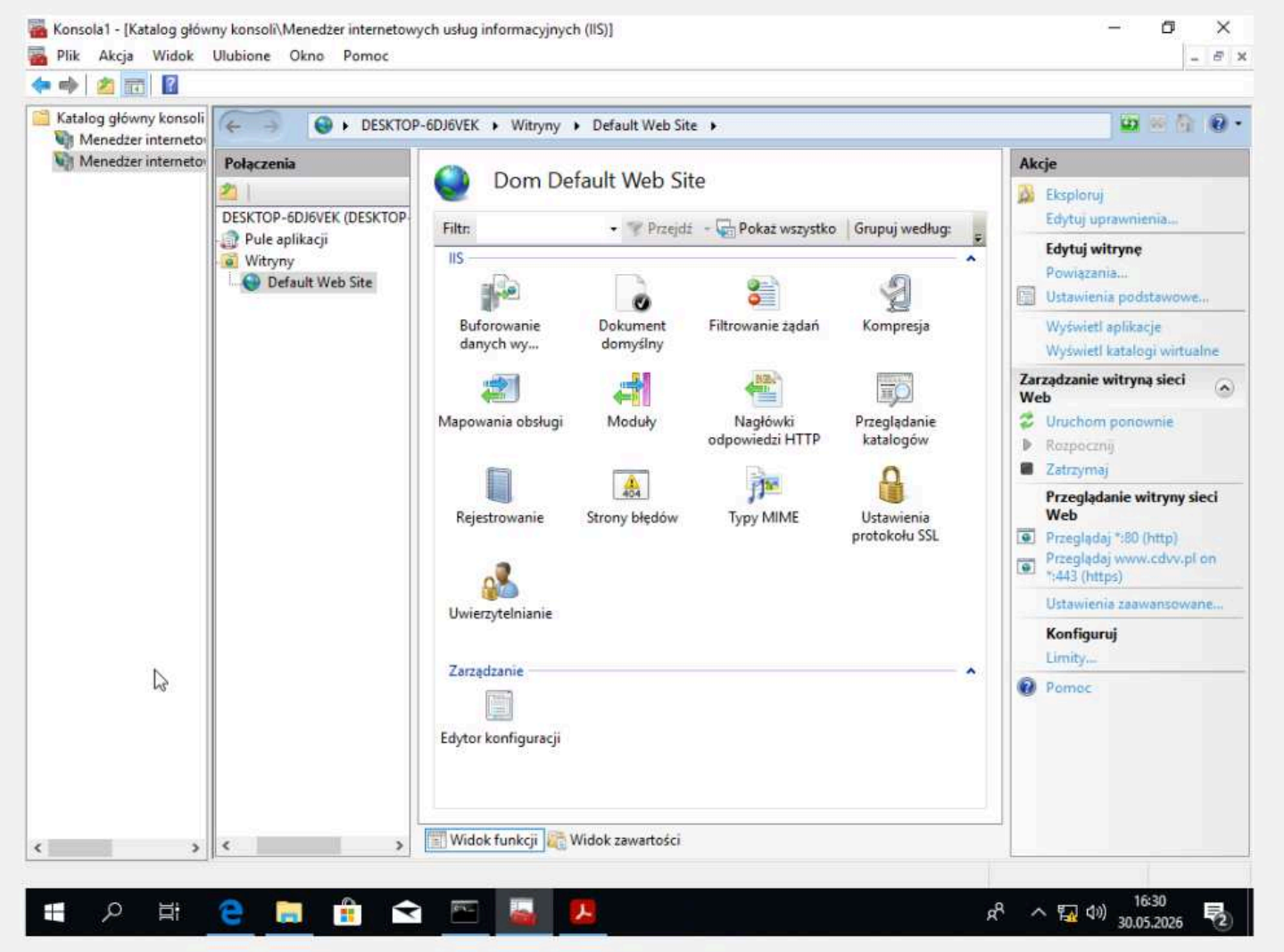


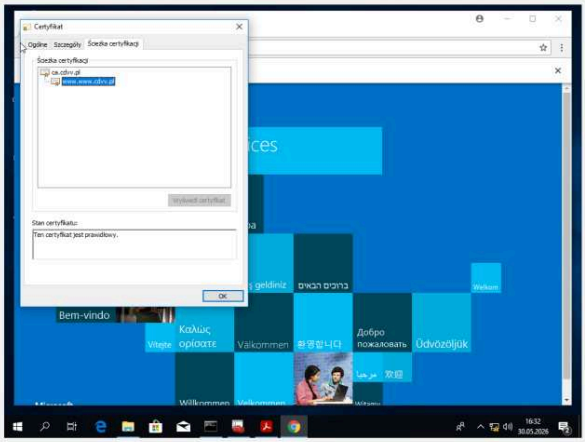
zmiana IP


Połączenia sieciowe
Panel sterowania › Sieć i Internet › Połączenia sieciowe
Przeszukaj: Połączenia sieciowe
Organizuj
Wyłącz to u
połączenia
Ethernet
ad.cdv.pl
Intel(R) PRO/
Właściwości: Ethernet 2
Sieć
Udostępnienie
Właściwości: Protokół internetowy w wersji 4 (TCP/IPv4)
Ogólne
Przy odpowiedniej konfiguracji sieci możesz automatycznie uzyskać niezbędne ustawienia protokołu IP. W przeciwnym wypadku musisz uzyskać ustawienia protokołu IP od administratora sieci.
Uzyskaj adres IP automatycznie
Użyj następującego adresu IP:
Adres IP:
192 . 168 . 10 . 19
Maska podsieci:
255 . 255 . 255 . 0
Brama domyślna:
Uzyskaj adres serwera DNS automatycznie
Użyj następujących adresów serwerów DNS:
Preferowany serwer DNS:
Alternatywny serwer DNS:
Sprawdź przy zakończeniu poprawność ustawień
Zaawansowane...
OK
Anuluj
Elementy: 2
1 zaznaczony element

Lab 2. OpenSSL - PKI

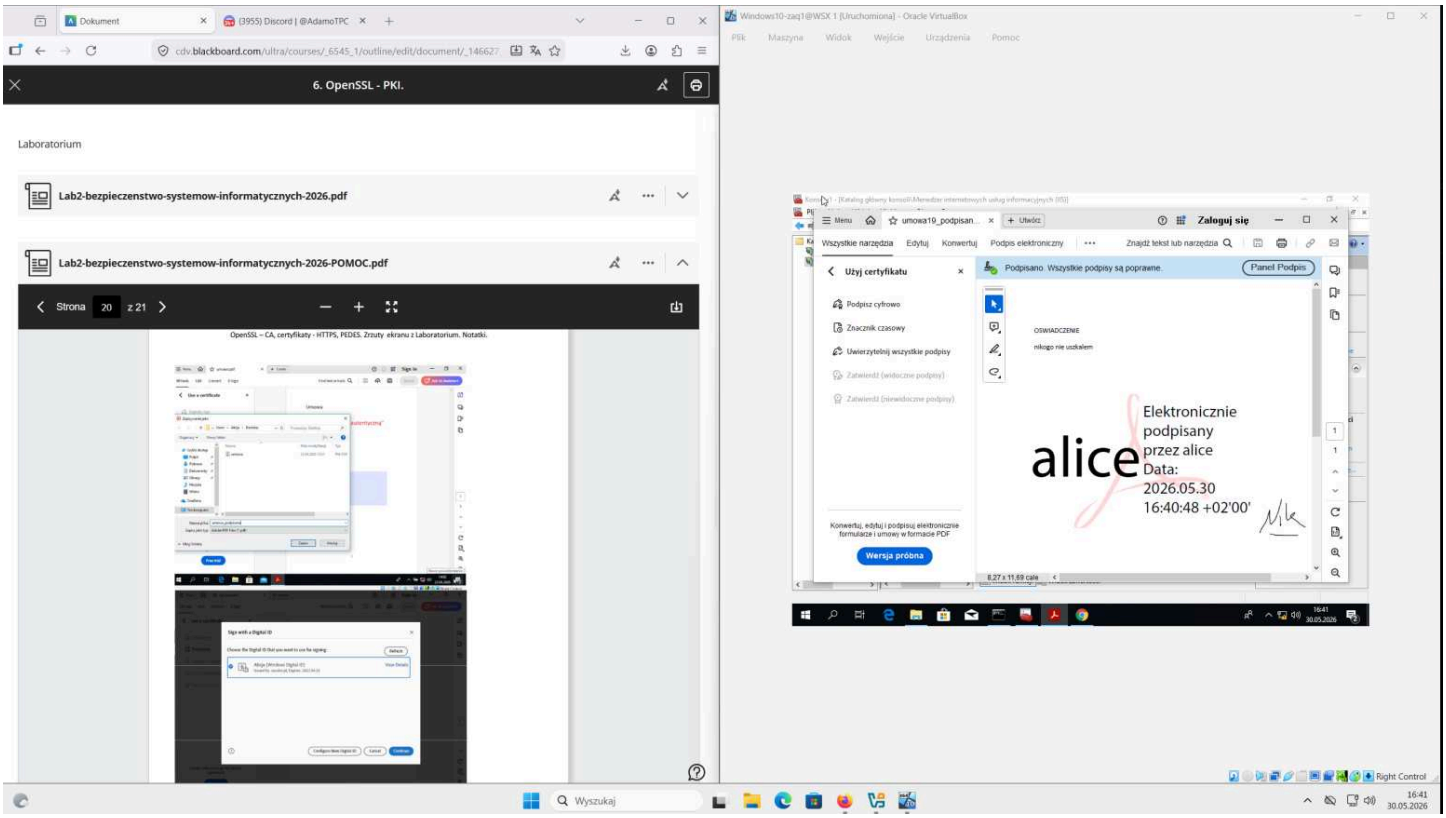
certyfikat

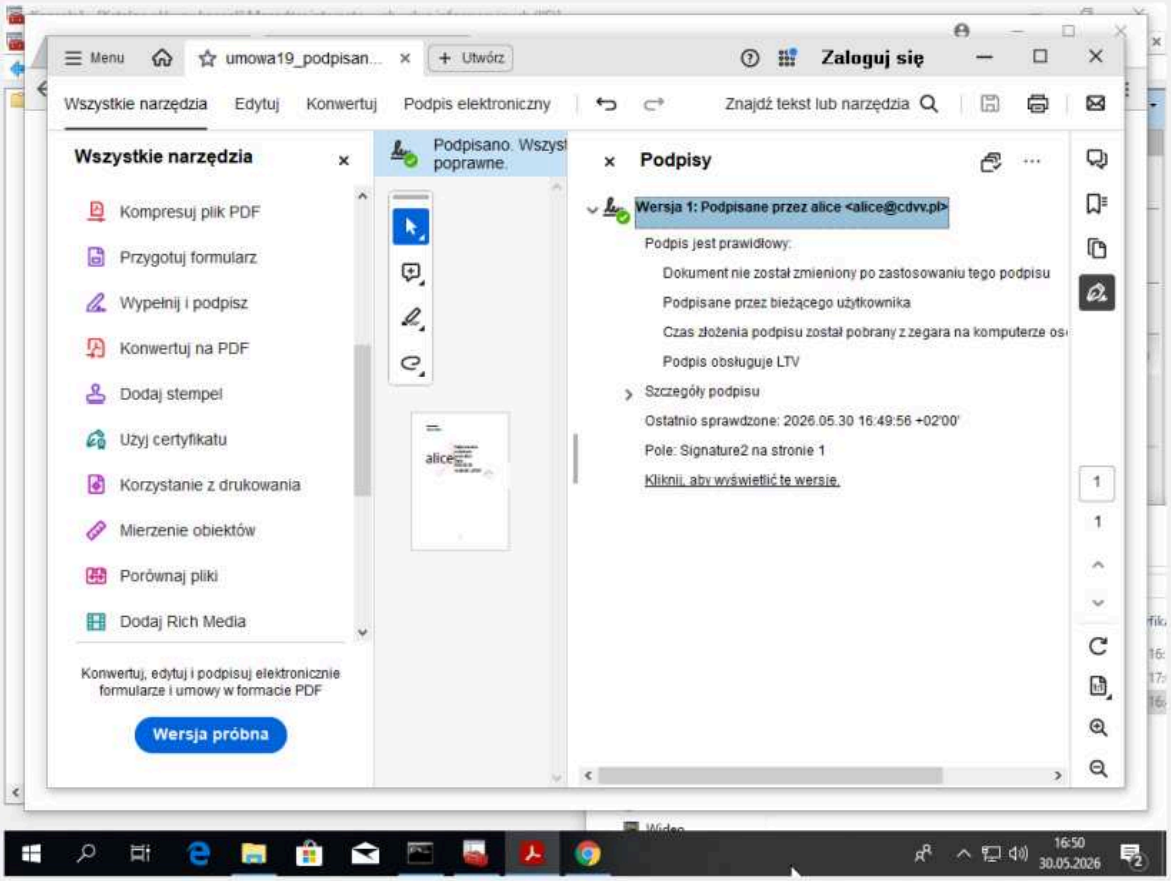



Certyfikat
Ogólne
Szczegóły
Ścieżka certyfikacji
Ścieżka certyfikacji
ca.cdrv.pl
www.www.cdrv.pl
Wyświetl certyfikat
Stan certyfikatu:
Ten certyfikat jest prawidłowy.
OK
Bem-vindo
Καλώς ορίσατε
Välkommen
환영합니다
Добро пожаловать
Üdvözöljük
Willkommen
Wyszukaj
Right Control
16:32
30.05.2026

Lab 2. OpenSSL - PKI

podpis

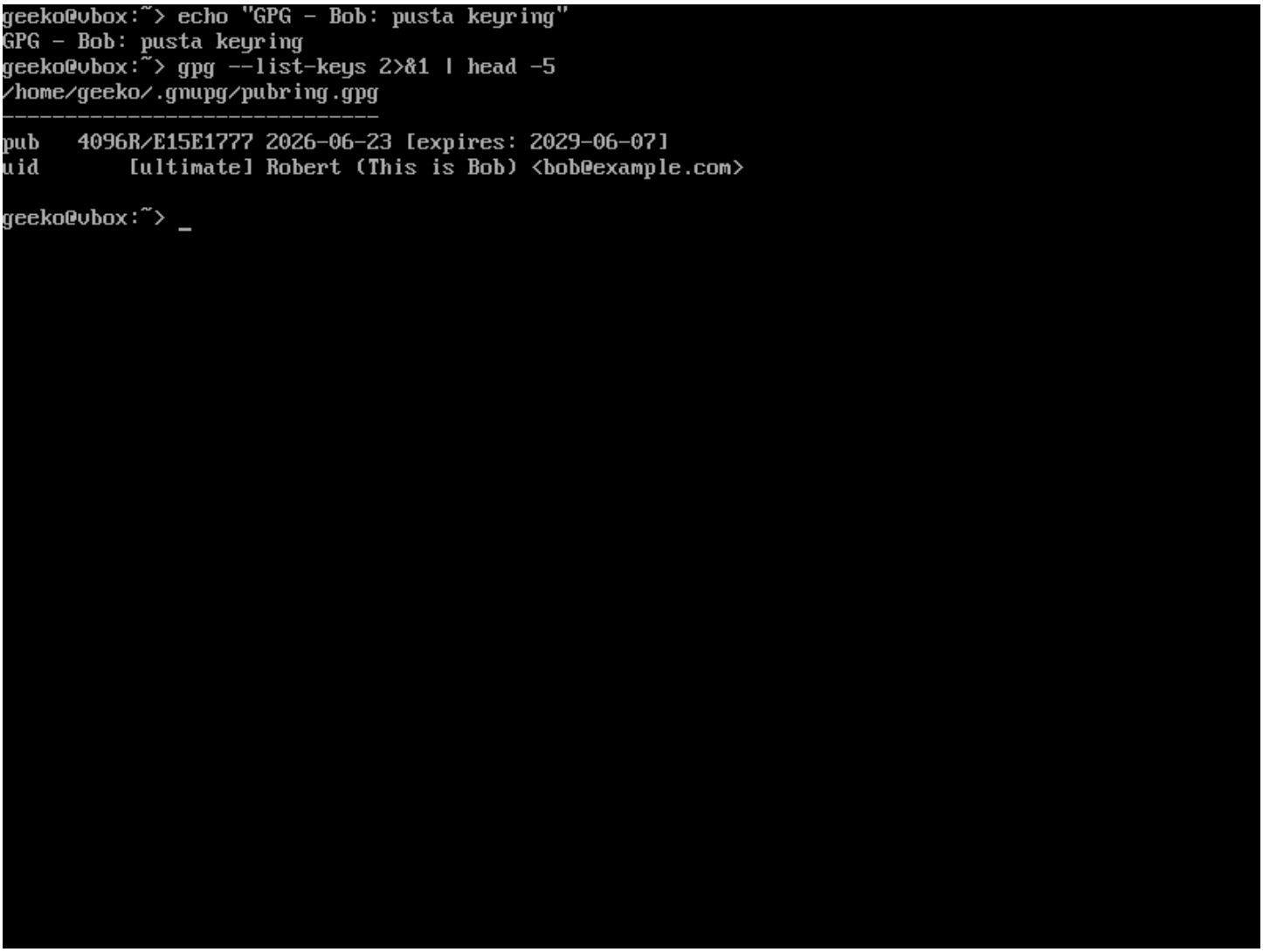




Right Control
16:50
30.05.2026

Lab 3. GPG.

(opisy w konsoli)



```
IC90bXAuYWxpY2UuYmF0Y2gKZmkKcm0gLWYgYWxpY2VfZ3Bn' >> /tmp/b64
linux-t5co:~ # printf '%s' 'LnB1YgpncGcgLS1hcm1vciAtbyBhbGljZV9ncGcucHViIC0tZXhwb3J0IGFsaWNlQGV4YW1w
bGUuY29tCnNjcCAtbyBTdHJpY3RIb3N0S2V5Q2hlY2tpbmc9' >> /tmp/b64
linux-t5co:~ # printf '%s' 'bm8gYWxpY2VfZ3BnLnB1YiAke0lQX1MxfToKZ3BnIC0tbGlzdC1rZXlzCg==' >> /tmp/b6
4
linux-t5co:~ # base64 -d /tmp/b64 > /tmp/run.sh && chmod +x /tmp/run.sh
linux-t5co:~ # bash /tmp/run.sh
gpg: keyring `/home/geeko/.gnupg/secring.gpg' created
gpg: key 698F67E2 marked as ultimately trusted
Password:
alice_gpg.pub                                                 100% 1653     1.6KB/s   00:00
gpg: checking the trustdb
gpg: 3 marginal(s) needed, 1 complete(s) needed, PGP trust model
gpg: depth: 0  valid:   1  signed:   0  trust: 0-, 0q, 0n, 0m, 0f, 1u
gpg: next trustdb check due at 2029-06-07
/home/geeko/.gnupg/pubring.gpg
------------------------------
pub   4096R/698F67E2 2026-06-23 [expires: 2029-06-07]
uid       [ultimate] Alice (This is Alice) <alice@example.com>

linux-t5co:~ # linux
If 'linux' is not a typo you can use command-not-found to lookup the package that contains it, like
this:
    cnf linux
linux-t5co:~ # gpg --list-keys
/root/.gnupg/pubring.gpg
------------------------
pub   2048R/39DB7C82 2013-01-31 [expired: 2017-01-30]
uid       [ expired] SuSE Package Signing Key <build@suse.de>

pub   2048R/50A3DD1C 2013-01-14 [expired: 2017-01-13]
uid       [ expired] SuSE Package Signing Key (reserve key) <build@suse.de>

pub   4096R/6B59C01E 2026-06-23 [expires: 2029-06-07]
uid       [ultimate] Alice (This is Alice) <alice@example.com>

linux-t5co:~ # _
```

```
     Robert (This is Bob) <bob@example.com>

This key is due to expire on 2029-06-07.
Are you sure that you want to sign this key with your
key "Alice (This is Alice) <alice@example.com>" (698F67E2)


gpg: checking the trustdb
gpg: 3 marginal(s) needed, 1 complete(s) needed, PGP trust model
gpg: depth: 0  valid:   1  signed:   1  trust: 0-, 0q, 0n, 0m, 0f, 1u
gpg: depth: 1  valid:   1  signed:   0  trust: 1-, 0q, 0n, 0m, 0f, 0u
gpg: next trustdb check due at 2029-06-07
/home/geeko/.gnupg/pubring.gpg
------------------------------
pub   4096R/698F67E2 2026-06-23 [expires: 2029-06-07]
uid       [ultimate] Alice (This is Alice) <alice@example.com>

pub   4096R/E15E1777 2026-06-23 [expires: 2029-06-07]
uid       [  full  ] Robert (This is Bob) <bob@example.com>

linux-t5co:~ # linux
If 'linux' is not a typo you can use command-not-found to lookup the package that contains it, like
this:
    cnf linux
linux-t5co:~ # gpg --list-keys
/root/.gnupg/pubring.gpg
------------------------
pub   2048R/39DB7C82 2013-01-31 [expired: 2017-01-30]
uid       [ expired] SuSE Package Signing Key <build@suse.de>

pub   2048R/50A3DD1C 2013-01-14 [expired: 2017-01-13]
uid       [ expired] SuSE Package Signing Key (reserve key) <build@suse.de>

pub   4096R/6B59C01E 2026-06-23 [expires: 2029-06-07]
uid       [ultimate] Alice (This is Alice) <alice@example.com>

linux-t5co:~ # _
```

```
geeko@linux-t5co:~> eval $(ssh-agent -s) >/dev/null
geeko@linux-t5co:~> ssh-add ~/.ssh/id_rsa
/home/geeko/.ssh/id_rsa: No such file or directory
geeko@linux-t5co:~> sles12
If 'sles12' is not a typo you can use command-not-found to lookup the package that contains it, like
 this:
    cnf sles12
geeko@linux-t5co:~> scp -o StrictHostKeyChecking=no 192.168.1.1:encrypt_to_alice.gpg 192.168.1.1:cle
artext.txt.asc .
linux
ls -l encrypt_to_alice.gpg cleartext.txt.asc
clear
echo "=== Odszyfrowanie wiadomosci od Boba ==="
gpg --batch --yes --passphrase 4lic3w0nd3rl4nd --decrypt encrypt_to_alice.gpg
geeko
linux
cd ~
clear
echo "=== Odszyfrowanie wiadomosci od Boba ==="
gpg --batch --yes --passphrase 4lic3w0nd3rl4nd --decrypt encrypt_to_alice.gpg
geeko
linux
cd ~
clear
echo "=== Odszyfrowanie wiadomosci od Boba ==="
gpg --batch --yes --passphrase 4lic3w0nd3rl4nd -o decrypted.txt --decrypt encrypt_to_alice.gpg && ec
ho "Tresc:" && cat decrypted.txt
ssh: connect to host 192.168.1.1 port 22: Connection timed out
geeko
linux
cd ~
gpg --batch --yes --passphrase 4lic3w0nd3rl4nd -o decrypted.txt --decrypt encrypt_to_alice.gpg
clear
echo "=== Odszyfrowanie wiadomosci od Boba ==="
ls -l encrypt_to_alice.gpg decrypted.txt
echo "Tresc odszyfrowana:"; cat decrypted.txt
_
```

```
geeko@linux-t5co:~> echo "=== Weryfikacja podpisu Boba ==="
=== Weryfikacja podpisu Boba ===
geeko@linux-t5co:~> gpg --verify cleartext.txt.asc
gpg: Signature made Tue 23 Jun 2026 01:36:13 PM CEST using RSA key ID E15E1777
gpg: Good signature from "Robert (This is Bob) <bob@example.com>" [full]
geeko@linux-t5co:~>
```

Lab 4. Hardening.

ping do s1, działa




geeko@vbox:~> echo "Hardening - icmp_echo_ignore_all=1"
Hardening - icmp_echo_ignore_all=1
geeko@vbox:~> echo linux | sudo -S sh -c "echo 1 > /proc/sys/net/ipv4/icmp_echo_ignore_all"
geeko@vbox:~> cat /proc/sys/net/ipv4/icmp_echo_ignore_all
1
geeko@vbox:~>

ping po



odrzuca słabe hasło

```
linux-t5co:~ # echo "Hardening - test polityki hasel (passwd)"
Hardening - test polityki hasel (passwd)
linux-t5co:~ # passwd
New password:
BAD PASSWORD: it is too short
BAD PASSWORD: is too simple
Retype new password:
Sorry, passwords do not match.
passwd: Failed preliminary check by password service
passwd: password unchanged
linux-t5co:~ # sles12
If 'sles12' is not a typo you can use command-not-found to lookup the package that contains it, like
 this:
    cnf sles12
linux-t5co:~ #
```

Lab 5. Firewall.

przed

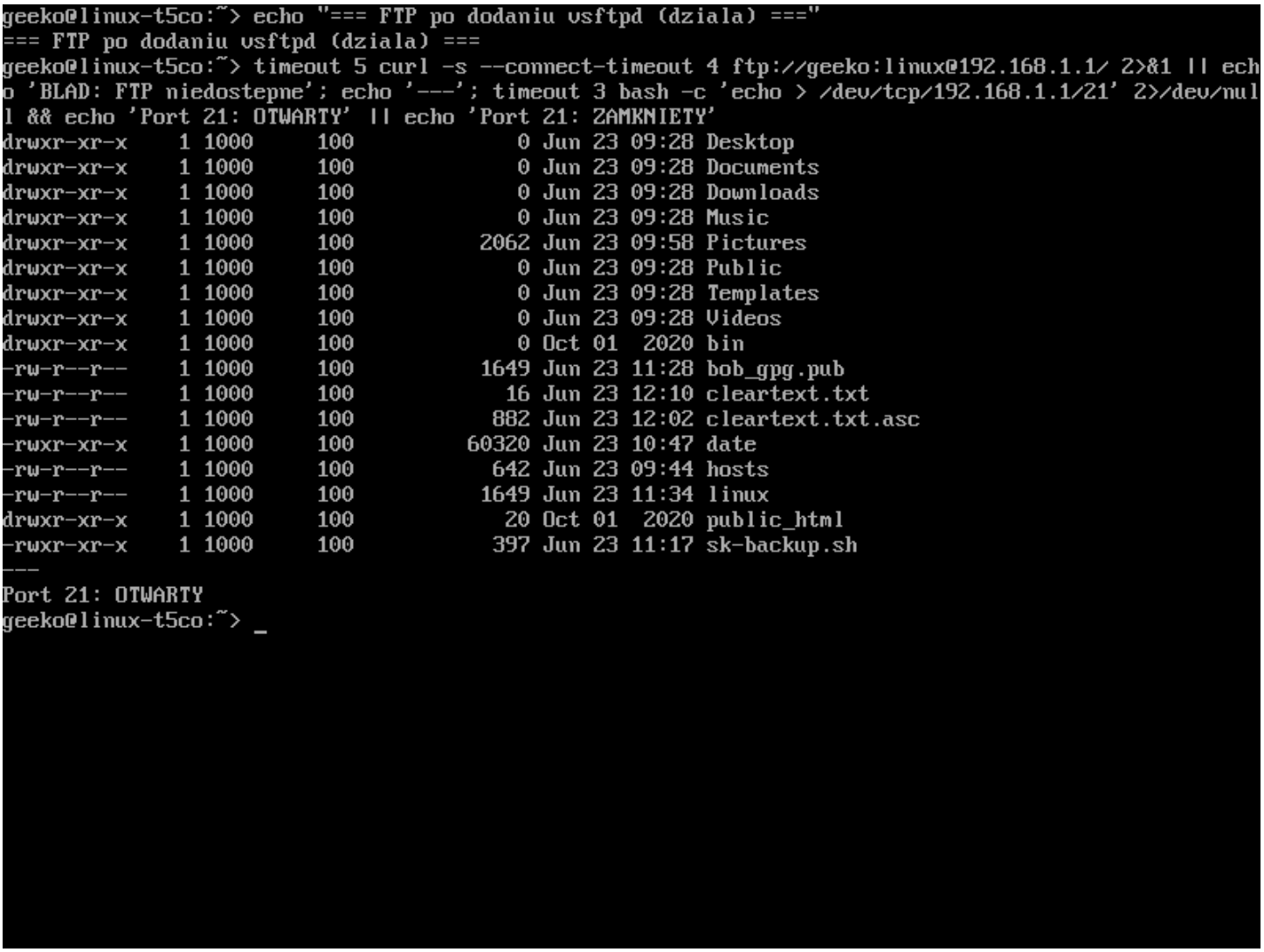
```
geeko@linux-t5co:~> echo "=== FTP przed firewallem ==="
=== FTP przed firewallem ===
geeko@linux-t5co:~> timeout 5 curl -s --connect-timeout 4 ftp://geeko:linux@192.168.1.1/ 2>&1; echo
'---'; timeout 3 bash -c 'echo > /dev/tcp/192.168.1.1/21' 2>/dev/null && echo 'Port 21: OTWARTY' ||
echo 'Port 21: ZAMKNIETY'
drwxr-xr-x    1 1000     100             0 Jun 23 09:28 Desktop
drwxr-xr-x    1 1000     100             0 Jun 23 09:28 Documents
drwxr-xr-x    1 1000     100             0 Jun 23 09:28 Downloads
drwxr-xr-x    1 1000     100             0 Jun 23 09:28 Music
drwxr-xr-x    1 1000     100          2062 Jun 23 09:58 Pictures
drwxr-xr-x    1 1000     100             0 Jun 23 09:28 Public
drwxr-xr-x    1 1000     100             0 Jun 23 09:28 Templates
drwxr-xr-x    1 1000     100             0 Jun 23 09:28 Videos
drwxr-xr-x    1 1000     100             0 Oct 01  2020 bin
-rw-r--r--    1 1000     100          1649 Jun 23 11:28 bob_gpg.pub
-rw-r--r--    1 1000     100            16 Jun 23 12:10 cleartext.txt
-rw-r--r--    1 1000     100           882 Jun 23 12:02 cleartext.txt.asc
-rwxr-xr-x    1 1000     100         60320 Jun 23 10:47 date
-rw-r--r--    1 1000     100           642 Jun 23 09:44 hosts
-rw-r--r--    1 1000     100          1649 Jun 23 11:34 linux
drwxr-xr-x    1 1000     100            20 Oct 01  2020 public_html
-rwxr-xr-x    1 1000     100           397 Jun 23 11:17 sk-backup.sh
---
Port 21: OTWARTY
geeko@linux-t5co:~> _
```

zablokowane przez fw


geeko@linux-t5co:~> echo "=== FTP po wlaczeniu firewall (zablokowane) ==="
=== FTP po wlaczeniu firewall (zablokowane) ===
geeko@linux-t5co:~> timeout 5 curl -s --connect-timeout 4 ftp://geeko:linux@192.168.1.1/ 2>&1 || ech
o 'BLAD: FTP niedostepne'; echo '---'; timeout 3 bash -c 'echo > /dev/tcp/192.168.1.1/21' 2>/dev/nul
l && echo 'Port 21: OTWARTY' || echo 'Port 21: ZAMKNIETY'
BLAD: FTP niedostepne

Port 21: ZAMKNIETY
geeko@linux-t5co:~> _

dziala



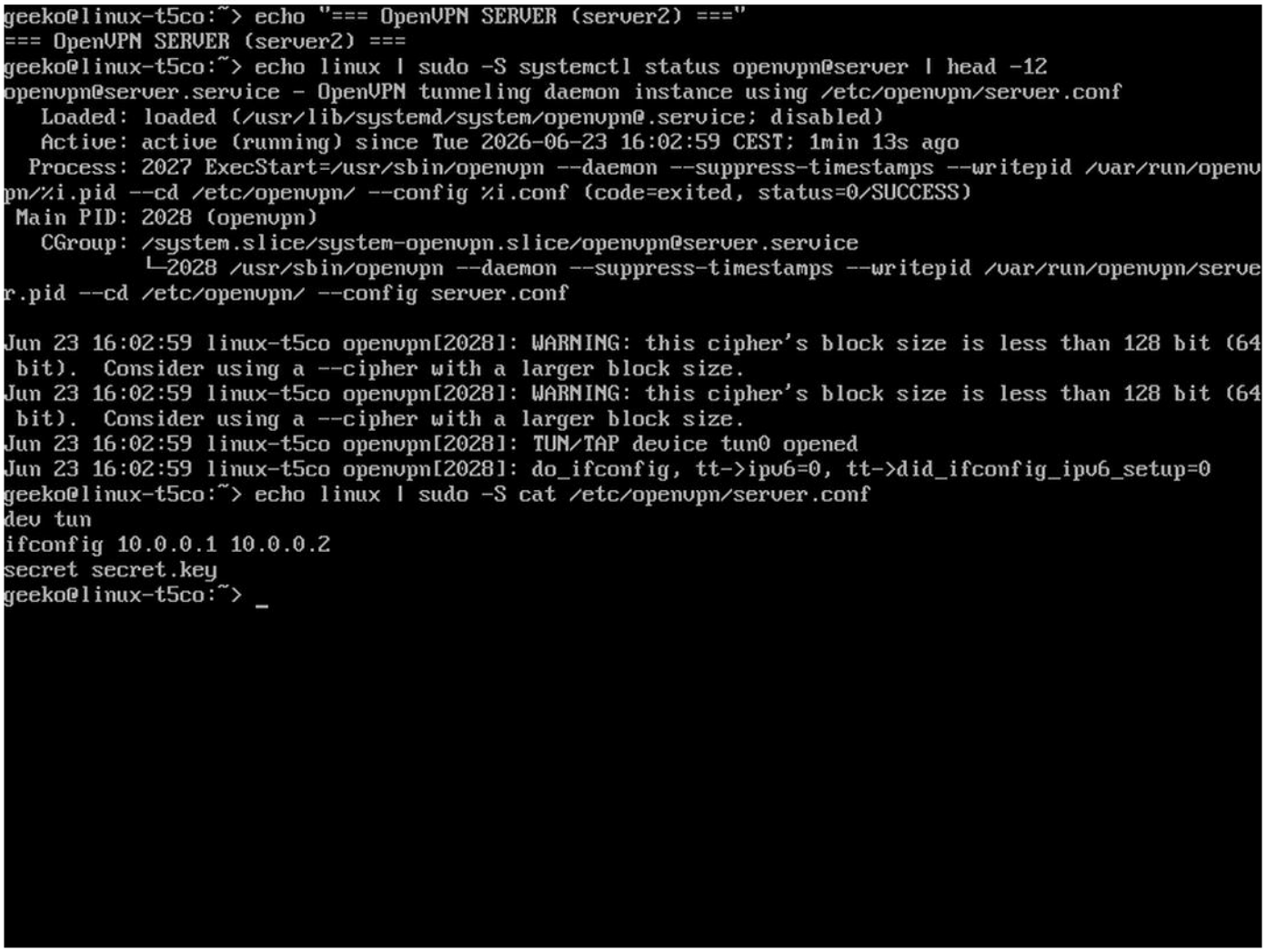
iptables, conf

```
Chain FORWARD (policy ACCEPT)
target     prot opt source               destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
geeko@vbox:~> grep -v "^#" /etc/sysconfig/SuSEfirewall2 | grep -v "^$" | grep -E "START_FW|FW_DEV|FW
_SERVICES"
FW_DEV_EXT="eth2"
FW_DEV_INT=""
FW_DEV_DMZ=""
FW_SERVICES_EXT_TCP="vsftpd"
FW_SERVICES_EXT_UDP=""
FW_SERVICES_EXT_IP=""
FW_SERVICES_EXT_RPC=""
FW_SERVICES_DMZ_TCP=""
FW_SERVICES_DMZ_UDP=""
FW_SERVICES_DMZ_IP=""
FW_SERVICES_DMZ_RPC=""
FW_SERVICES_INT_TCP=""
FW_SERVICES_INT_UDP=""
FW_SERVICES_INT_IP=""
FW_SERVICES_INT_RPC=""
FW_SERVICES_DROP_EXT=""
FW_SERVICES_DROP_DMZ=""
FW_SERVICES_DROP_INT=""
FW_SERVICES_REJECT_EXT=""
FW_SERVICES_REJECT_DMZ=""
FW_SERVICES_REJECT_INT=""
FW_SERVICES_ACCEPT_EXT=""
FW_SERVICES_ACCEPT_DMZ=""
FW_SERVICES_ACCEPT_INT=""
FW_SERVICES_ACCEPT_RELATED_EXT=""
FW_SERVICES_ACCEPT_RELATED_DMZ=""
FW_SERVICES_ACCEPT_RELATED_INT=""
START_FW="yes"
FW_SERVICES_EXT="vsftpd"
geeko@vbox:~> _
```

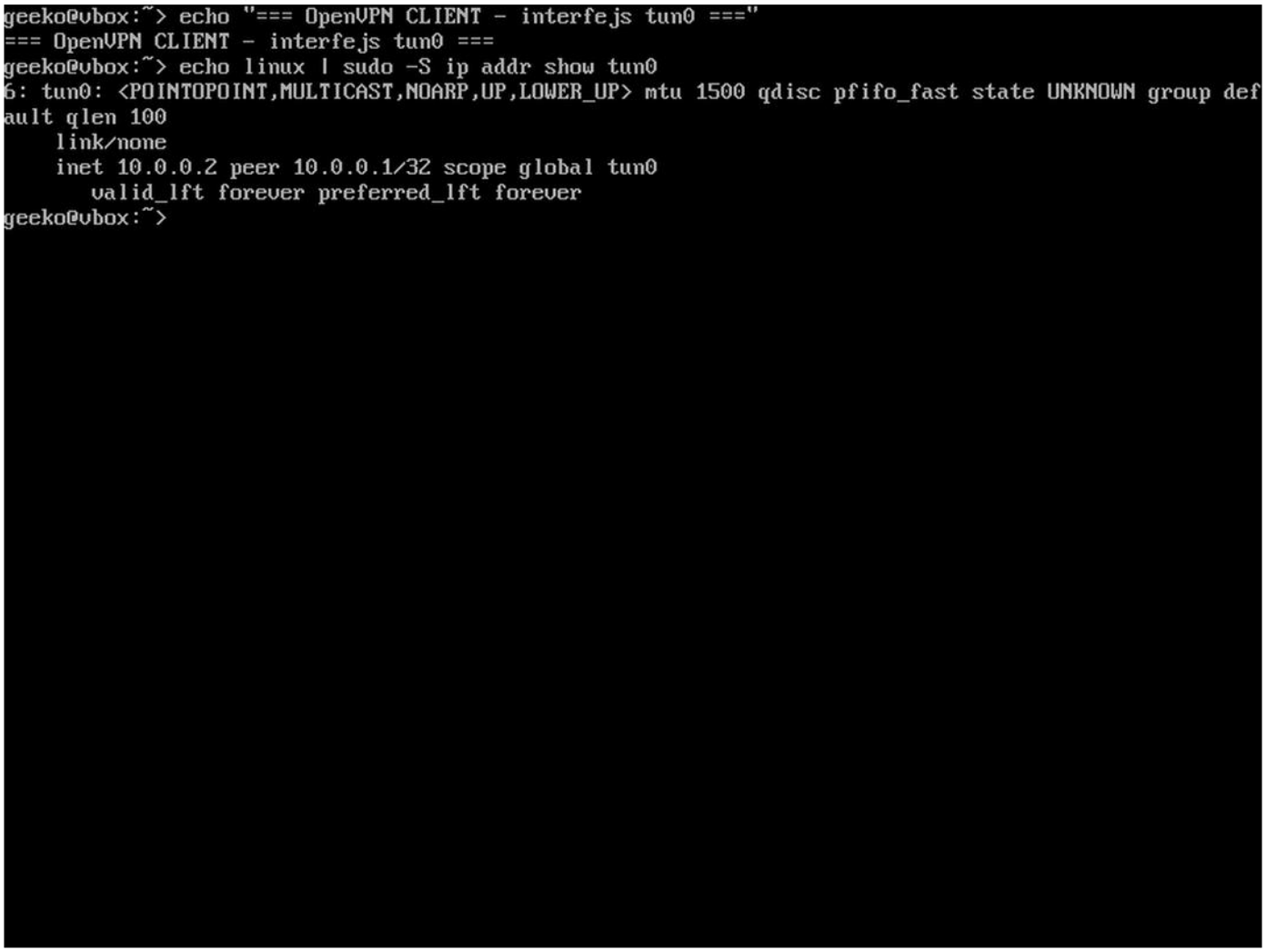
Lab 6. OpenVPN - konfiguracja serwera

screenshots/openvpn/openvpn01_server_setup.png



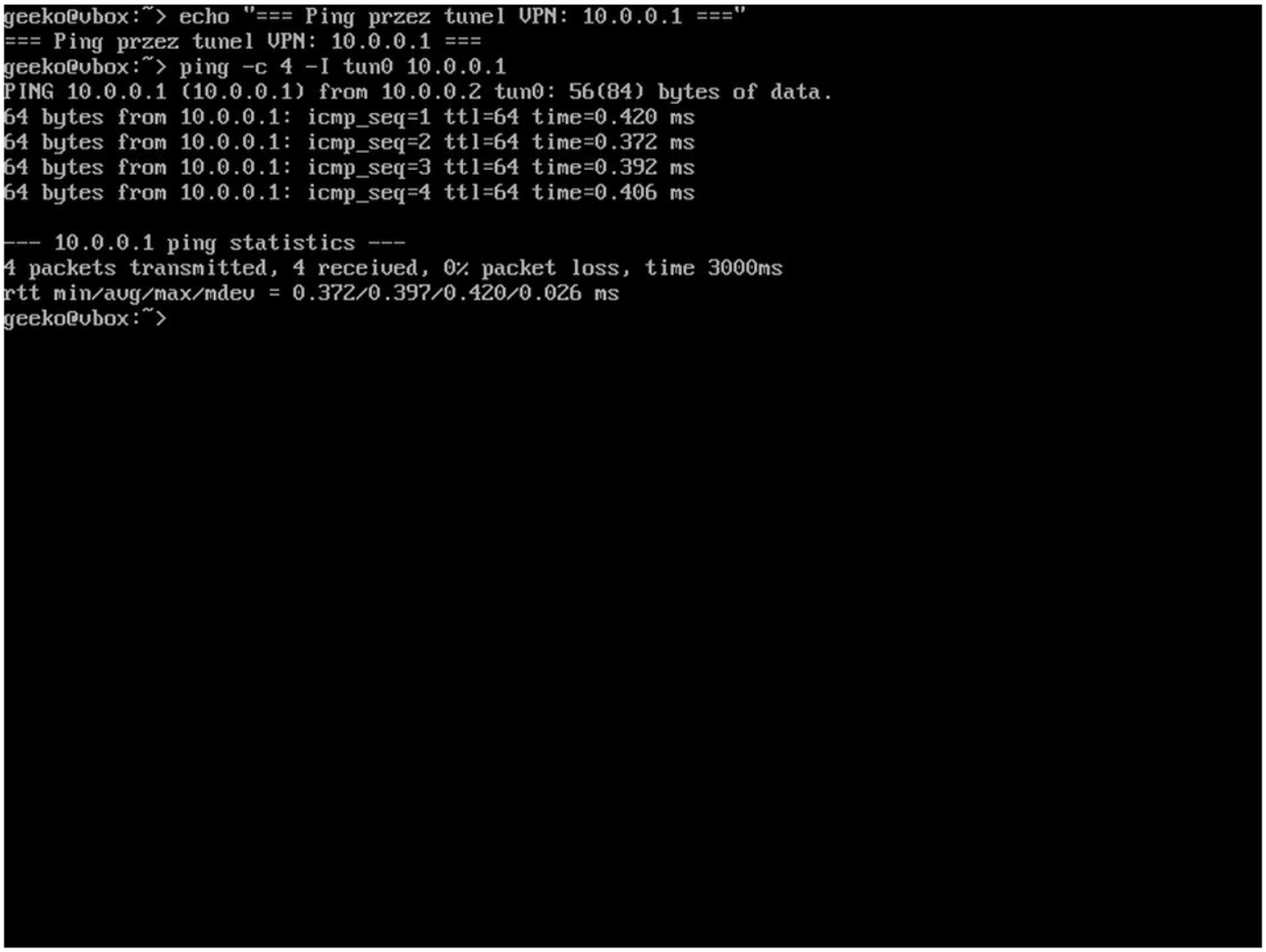
Lab 6. OpenVPN - uruchomiony tunel VPN (klient)

screenshots/openvpn/openvpn02_tunnel_up.png



Lab 6. OpenVPN - ping przez tunel VPN

screenshots/openvpn/openvpn03_ping_tunnel.png



Lab 6. OpenVPN - serwer: tun0 i log polaczenia

screenshots/openvpn/openvpn04_server_connected.png

